



AKTU



B. Tech II-Year

Digital Electronics

Unit-5 in One Shot

Pdf Notes + AKTU PYQs + Imp. Questions



By Swati Ma'am



AKTU B.Tech II-Year

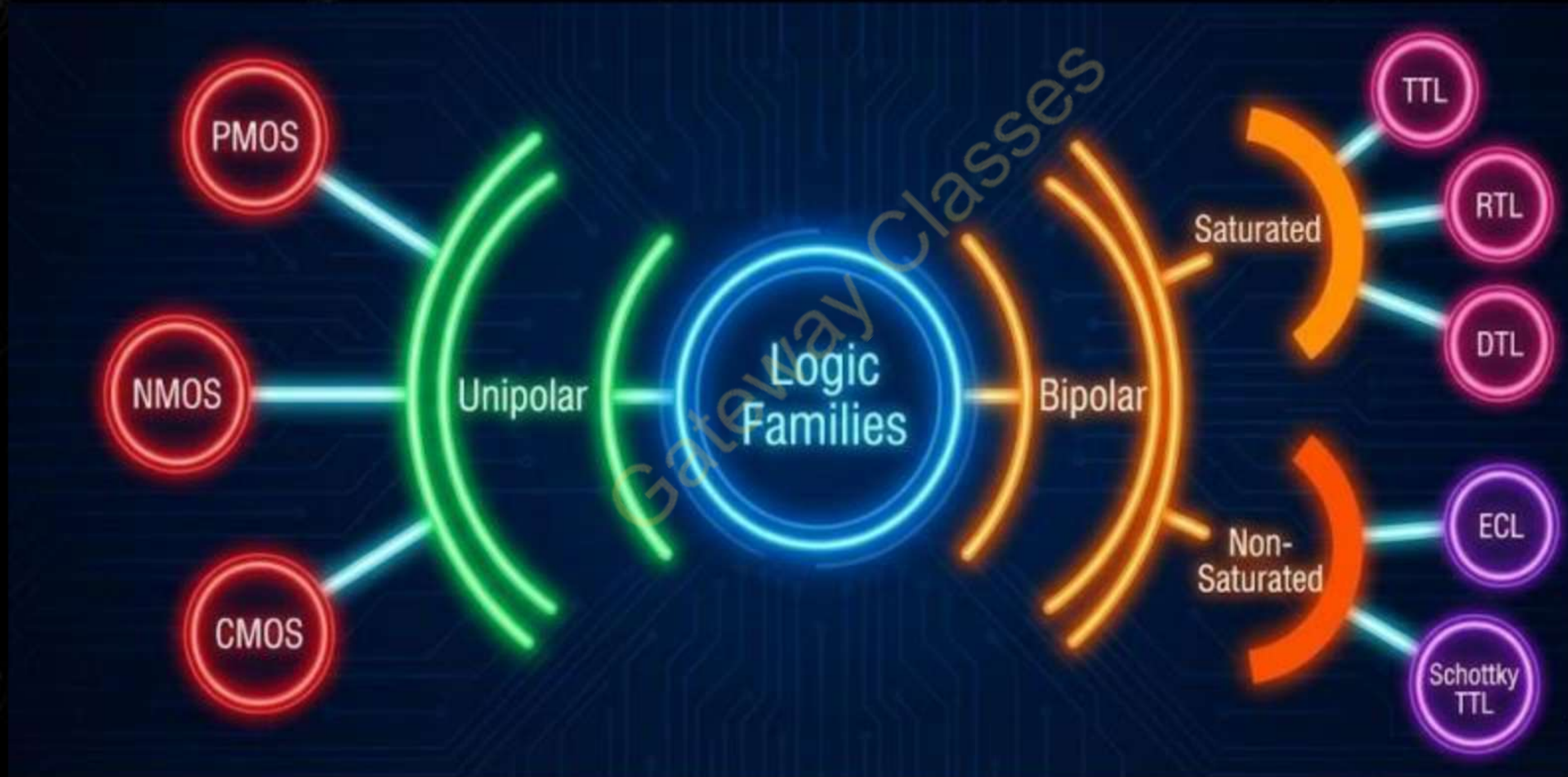
Subject : DIGITAL ELECTRONICS



Unit-5 : Syllabus

Memory & Programmable Logic Devices: Digital Logic Families: DTL, DCTL, TTL, ECL & CMOS etc., Fan Out, Fan in, Noise Margin; RAM, ROM, PLA, PAL; Circuits of Logic Families, Interfacing of Digital Logic Families, Circuit Implementation using ROM, PLA and PAL.

It is a group of compatible(same) ICs(Integrated Circuits) with the same logic levels and supply voltages fabricated for performing various logical functions.



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Logic families (like TTL, CMOS, ECL) are used as **fundamental building blocks** in almost all digital systems, enabling complex functions in **computers, calculators, control systems, and mobile devices**, with specific families chosen for trade-offs like speed (ECL), low power (CMOS for battery devices), or general purpose use (TTL).

1. Propagation Delay :- It is the time interval between the application of an input signal and the change in the output signal.

It is also called the "reaction time" of the logic gate as it measures how fast a signal travels through the Logic Gate. It is measured in nanoseconds (n_s) or picoseconds (p_s).

It is usually defined by two specific metrics:

t_{PHL} : Time taken for the output to switch from High to Low.

t_{PLH} : Time taken for the output to switch from Low to High.

$$\text{Average Delay: } t_{pd} = (t_{PHL} + t_{PLH})/2$$

It is the average delay time for the signal to propagate from input to output.

Significance: This limits the maximum speed (frequency) at which a digital system can run. If the delay is too long, the next stage of the circuit won't receive the data in time.

2. Fan-in : It is the maximum number of inputs that can be applied to a Logic Gate.

Example: A 2-input NAND gate has a fan-in of 2; an 8-input NAND gate has a fan-in of 8.

3. Fan-out It is the maximum number of loads that the output of a Logic Gate can drive.

4. Power Dissipation :- It is the power consumed per Logic Gate. It is expressed in milliwatts or nanowatts.

5. Figure of Merit :- It is the product of Propagation Delay and Power Dissipation.

(FOM)

6. Logic Swing :- It is the difference between two output voltages.

$$\text{Logic Swing} = V_{OH} - V_{OL}$$

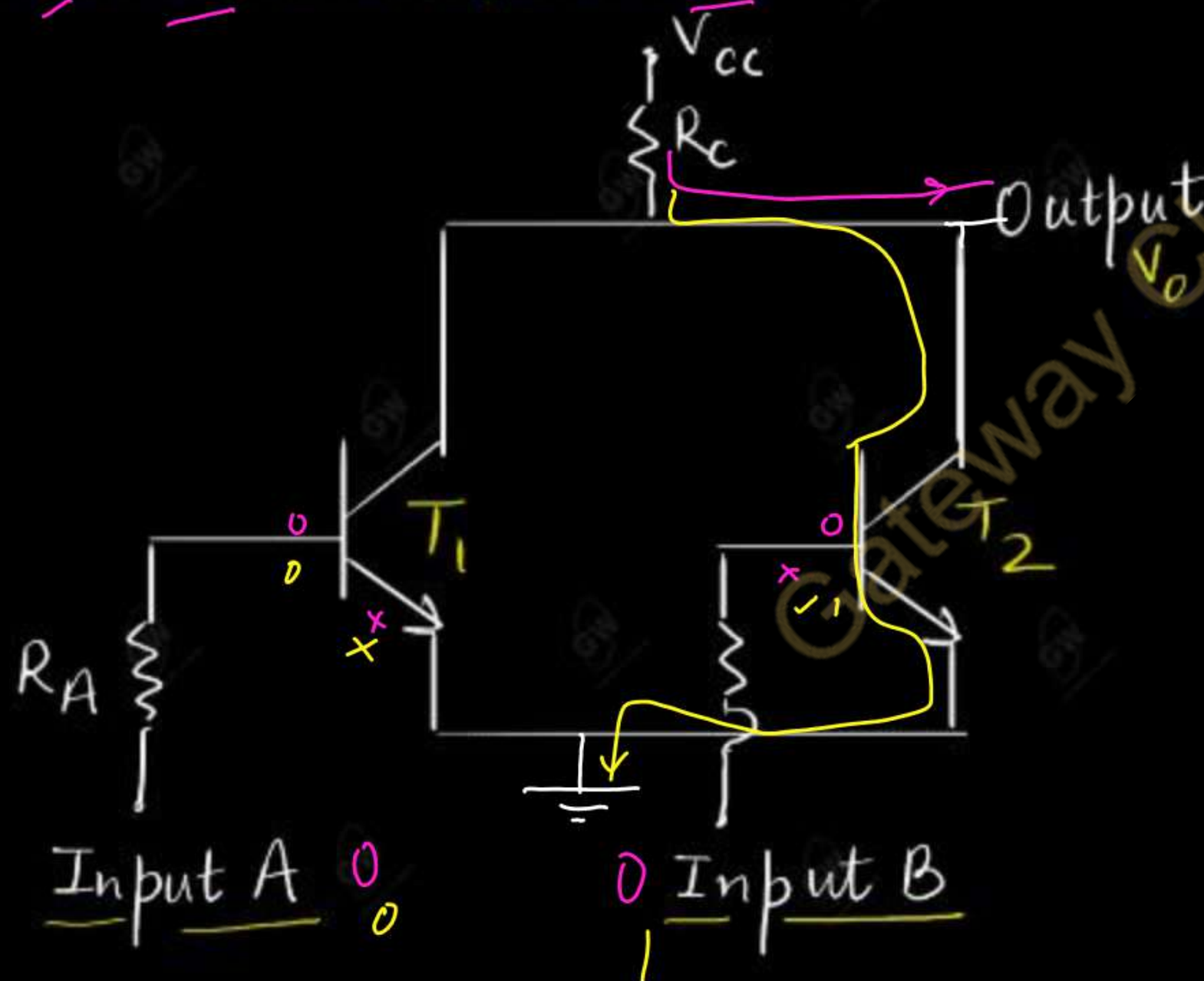
0 → V_{OL}
1 → V_{OH}

7. Noise Margin (Immunity):- It is the maximum noise that a circuit can withstand without changing the output.

Resistor-Transistor Logic (RTL) as NOR Gate

It has 2 inputs A,B.

✓ If 1 at Base then T1 ,T2 are ON.

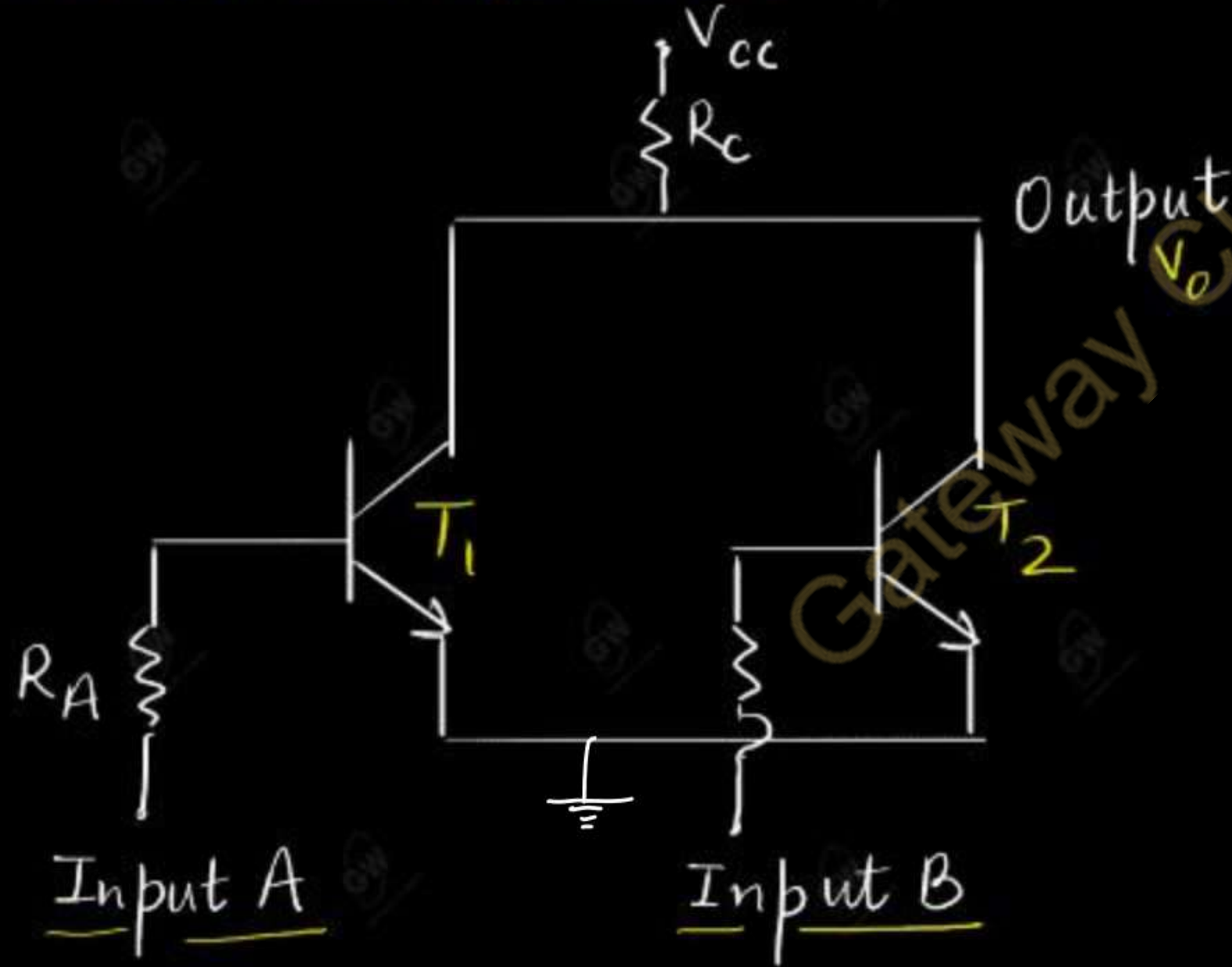


A	B	T1	T2	VO
0	0	OFF	OFF	1
0	1	OFF	ON	0
1	0	ON	OFF	0
1	1	ON	ON	0

Resistor-Transistor Logic (RTL) as NOR Gate

It has 2 inputs A,B.

✓ If 1 at Base then T1 ,T2 are ON.



A	B	T1	T2	VO
0	0	OFF	OFF	1
0	1	OFF	ON	0
1	0	ON	OFF	0
1	1	ON	ON	0

DTL (Diode Transistor Logic)

Low Level = 0.2 V (0)
High Level = 4.5 V (1)
NAND Gate

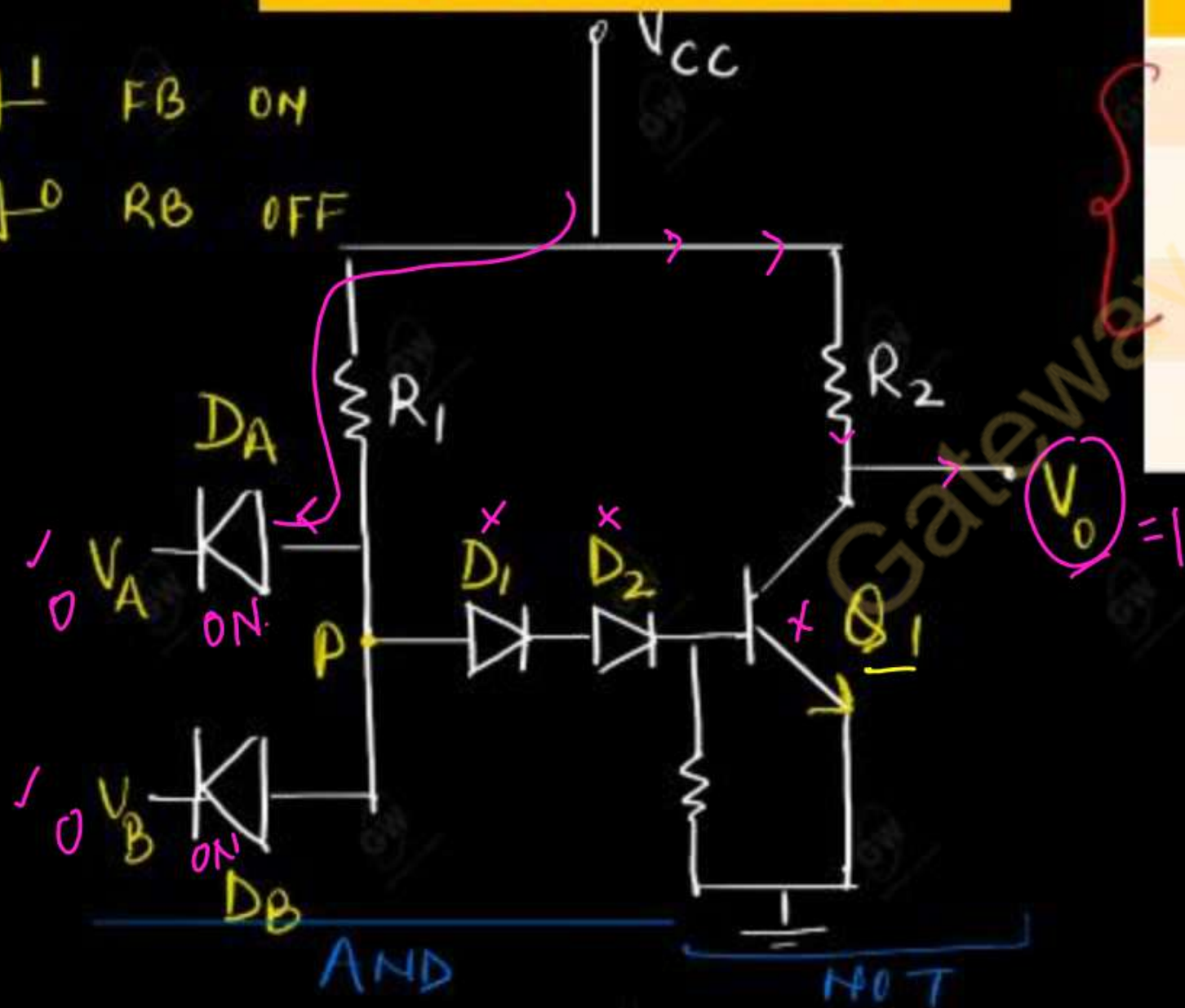
In DTL, diodes are used in the input stage, which cause large delay due to diode storage time.

- Saturated logic Family
- Better than RTL
- Higher Fanout than RTL
- Low Power Dissipation
- Low Speed

DTL NAND

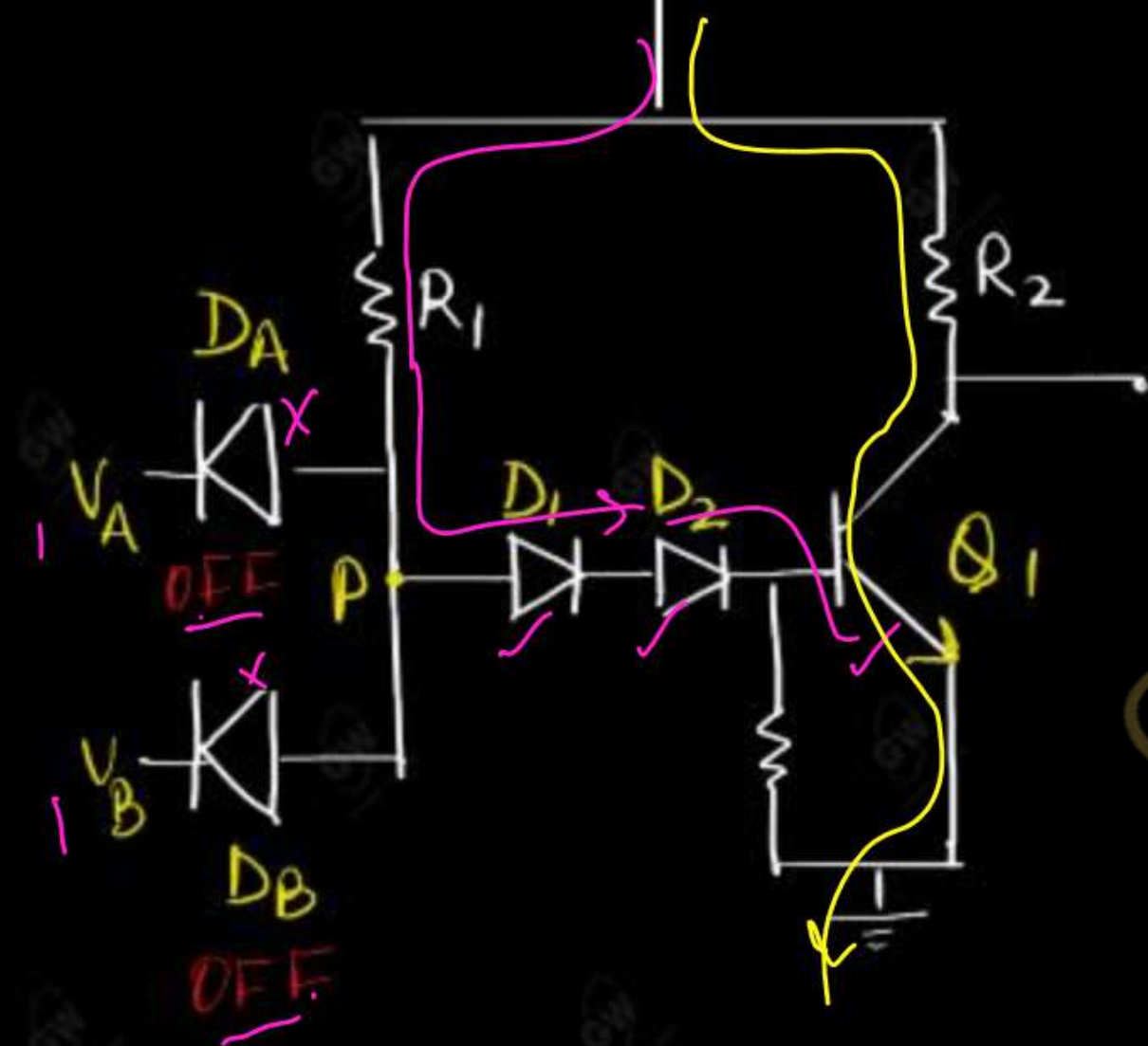
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V_A	V_B	D_A	D_B	D_1	D_2	Q_1	V_O
0	0	ON	ON	OFF	OFF	OFF	H
0	1	ON	OFF	OFF	OFF	OFF	H
1	0	OFF	ON	OFF	OFF	OFF	H
1	1	OFF	OFF	ON	ON	ON	L



0 $\rightarrow$ 1 FB ON

1 $\rightarrow$ 0 RB OFF V_{CC}



V_A	V_B	D_A	D_B	D_1	D_2	Q_1	V_O
0	0	ON	ON	OFF	OFF	OFF	H
0	1	ON	OFF	OFF	OFF	OFF	H
1	0	OFF	ON	OFF	OFF	OFF	H
1	1	OFF	OFF	ON	ON	ON	L

Transistor-Transistor Logic (TTL)

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In DTL, diodes are used in the input stage, which cause large delay due to diode storage time. TTL uses transistors instead of diodes, resulting in much faster switching.

The Transistor-Transistor Logic (TTL) is a logic family made up of BJTs (bipolar junction transistors).

The best examples of TTL are logic gates namely the 7402 NOR Gate & the 7400 NAND gate.

It also offers Higher Noise Immunity, Better Fan-out Capability, Improved Temperature Stability, $FOM = 100$

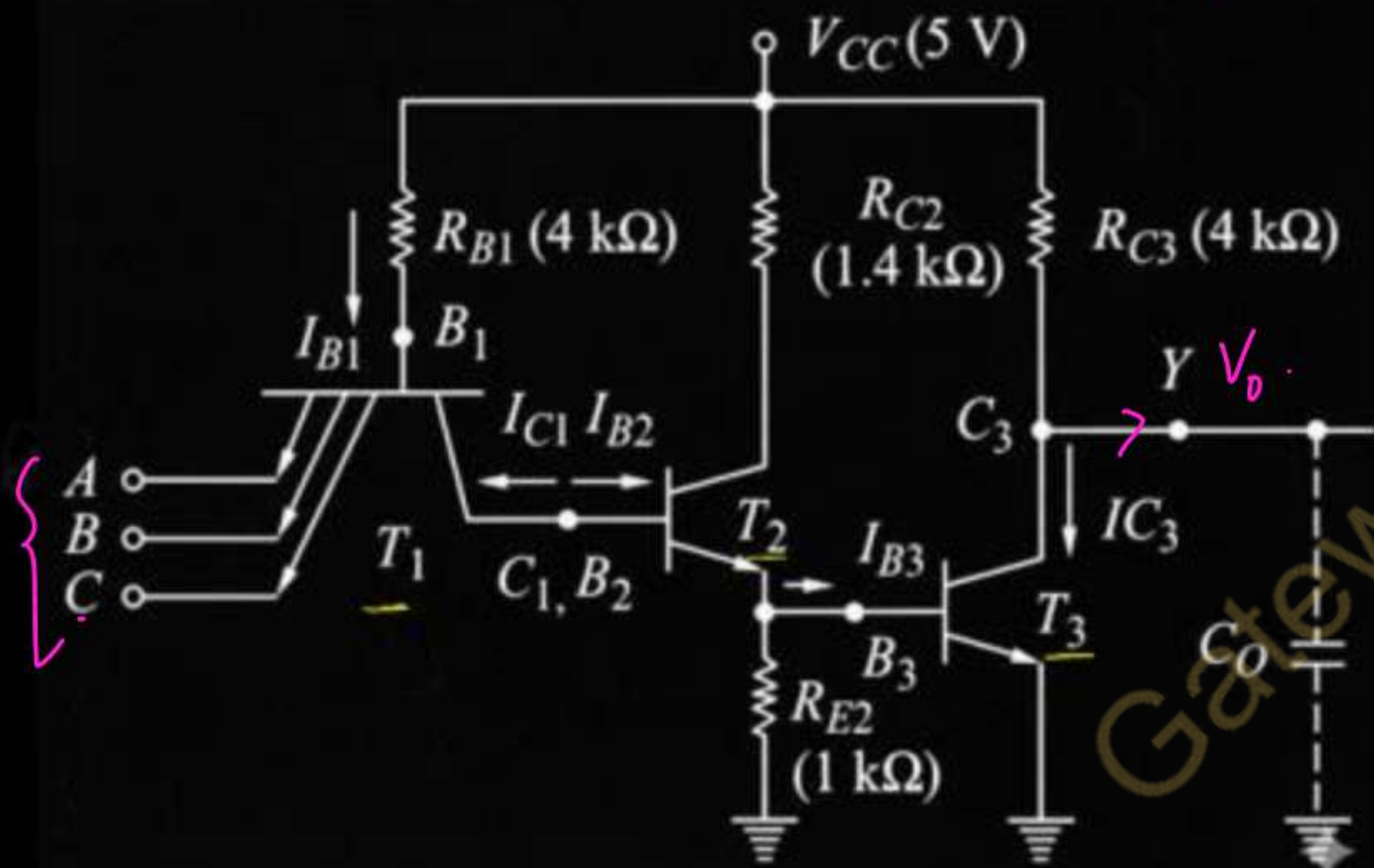
 $\rightarrow 0.4$ $\rightarrow 10$

It has 3 inputs A, B, C

T1 will be ON if A=B=C=1 otherwise OFF.
If 1 at Base then T2, T3 are ON.

✓ Base 1 T ON

T₁ driven T₂ driven T₃



A	B	C	T1	T2	T3	V0
0	0	0	OFF	OFF	OFF	1
0	0	1	OFF	OFF	OFF	1
0	1	0	OFF	OFF	OFF	1
0	1	1	OFF	OFF	OFF	1
1	0	0	OFF	OFF	OFF	1
1	0	1	OFF	OFF	OFF	1
1	1	0	OFF	OFF	OFF	1
1	1	1	ON	ON	ON	0

The TTL family uses transistors operating in the saturation mode. As a result, their switching speed is limited by the storage delay time associated with a transistor that is driven into saturation.

Therefore another logic family called Emitter-Coupled Logic (ECL) was developed.

ECL is the **fastest** among all logic families because in ECL Transistors are deliberately prevented from entering complete saturation, which avoids storage delays and allows very fast switching and so ECL have propagation delays around 1 ns.

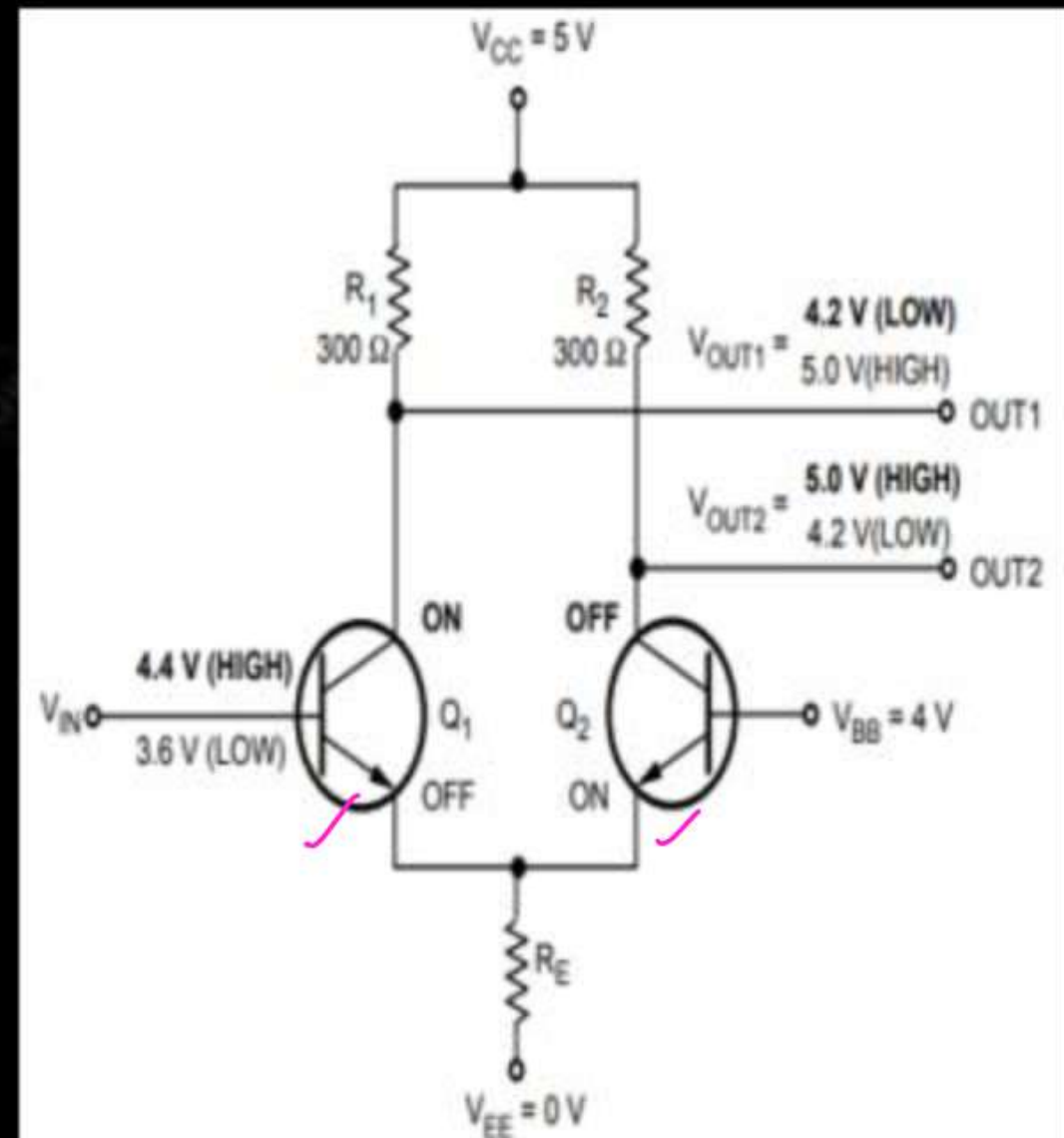


Fig. 2.7.1 Basic ECL inverter/buffer circuit

MOS Families

→ MOSFET

Digital circuits with MOSFETs can be grouped into three categories :

• PMOS

uses only P-channel

Enhancement MOSFETs



NMOS

uses only N-channel

Enhancement MOSFETs



CMOS

(Complementary MOS)

uses both

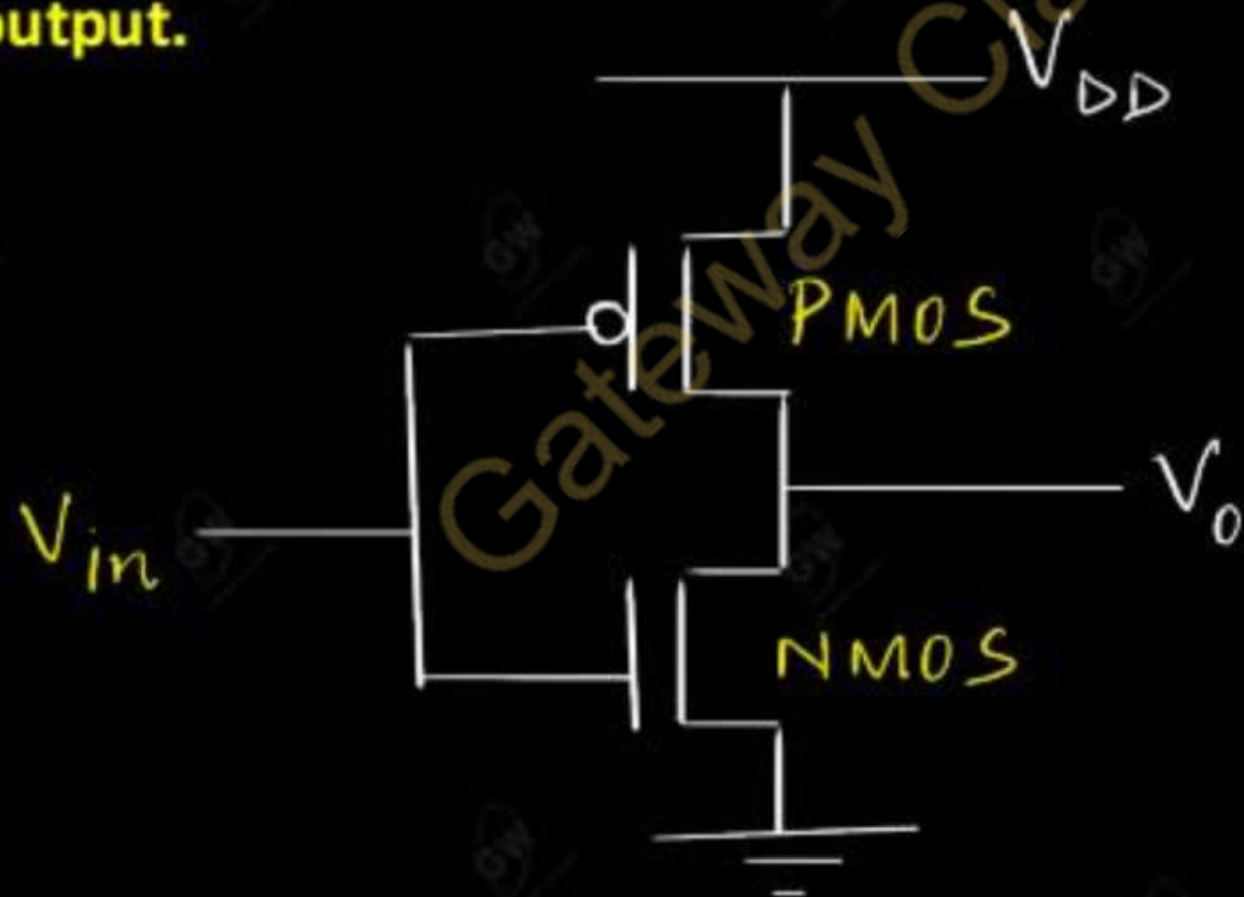
P and N-channel devices.

CMOS Inverter

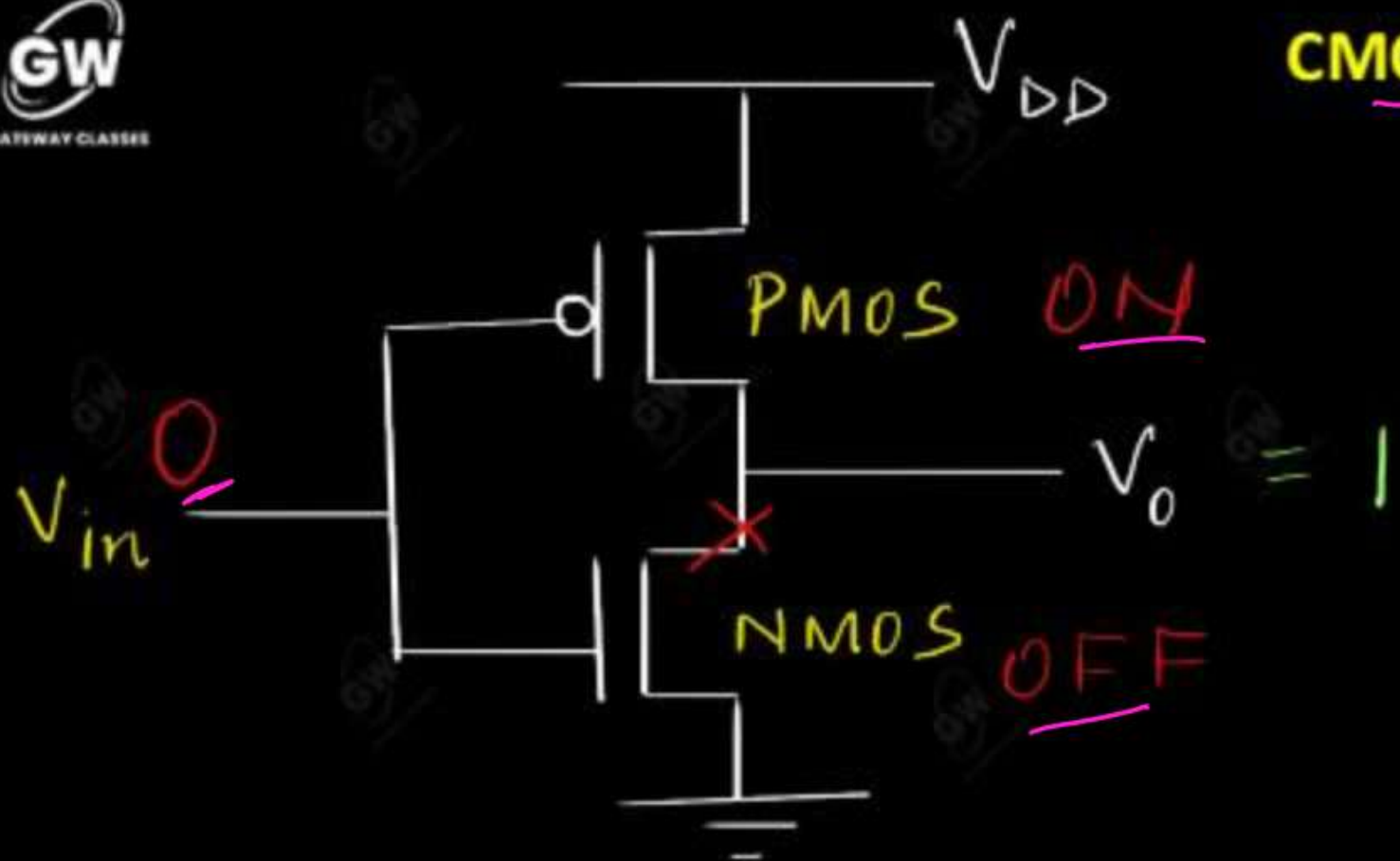
NOT

It consists of two MOSFETs Q1 and Q2 connected in series in such a way that the **p**-channel device has its source connected to **$+V_{DD}$** (a positive voltage) and the **n**-channel device has its source connected to **ground**.

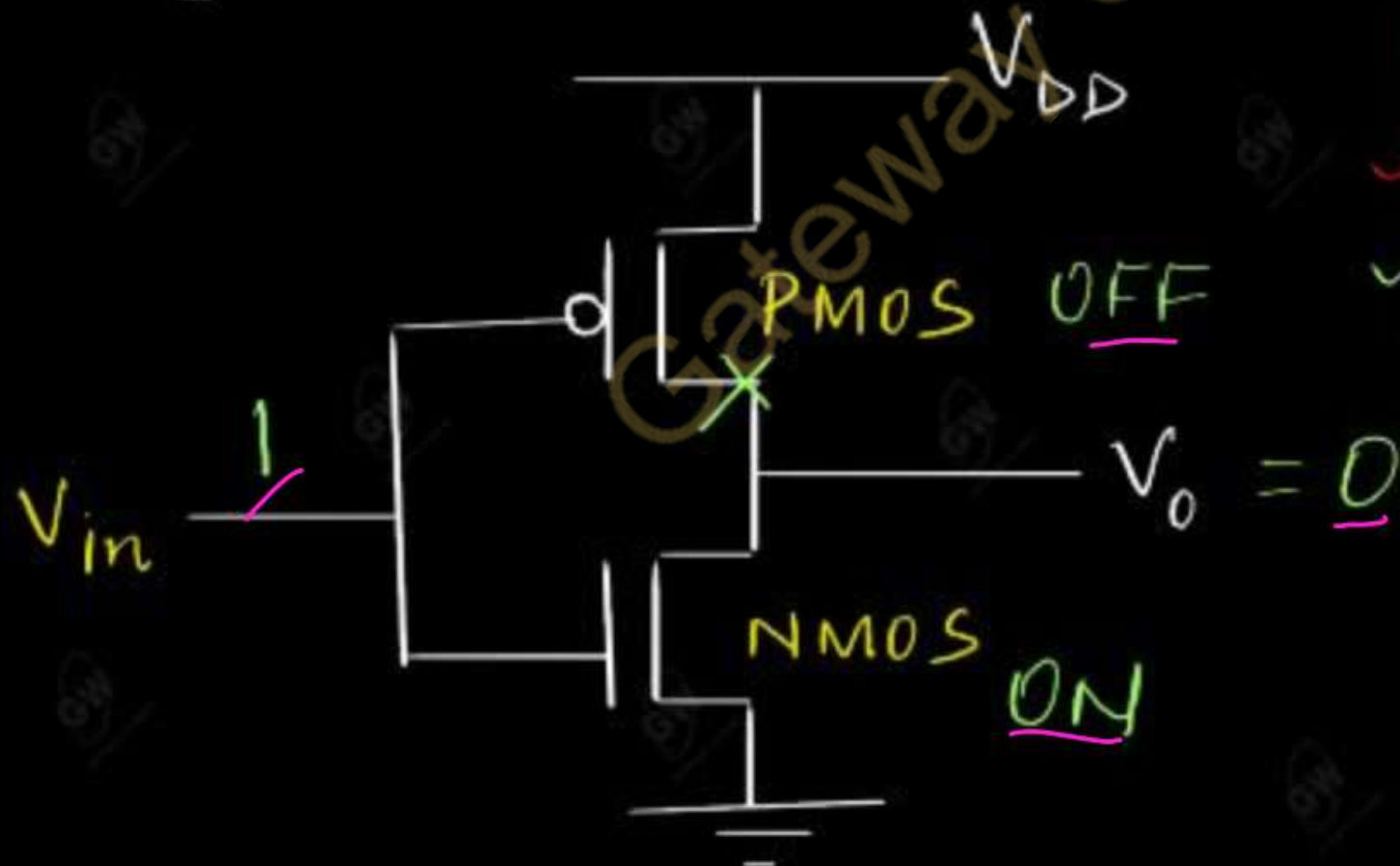
The **Gates** of the two devices are connected together as the **common input** and the **Drains** are connected together as the **common output**.



CMOS Inverter



When	$V_{in} = 0$	PMOS ON	NMOS OFF
	$V_{in} = 1$	PMOS OFF	NMOS ON

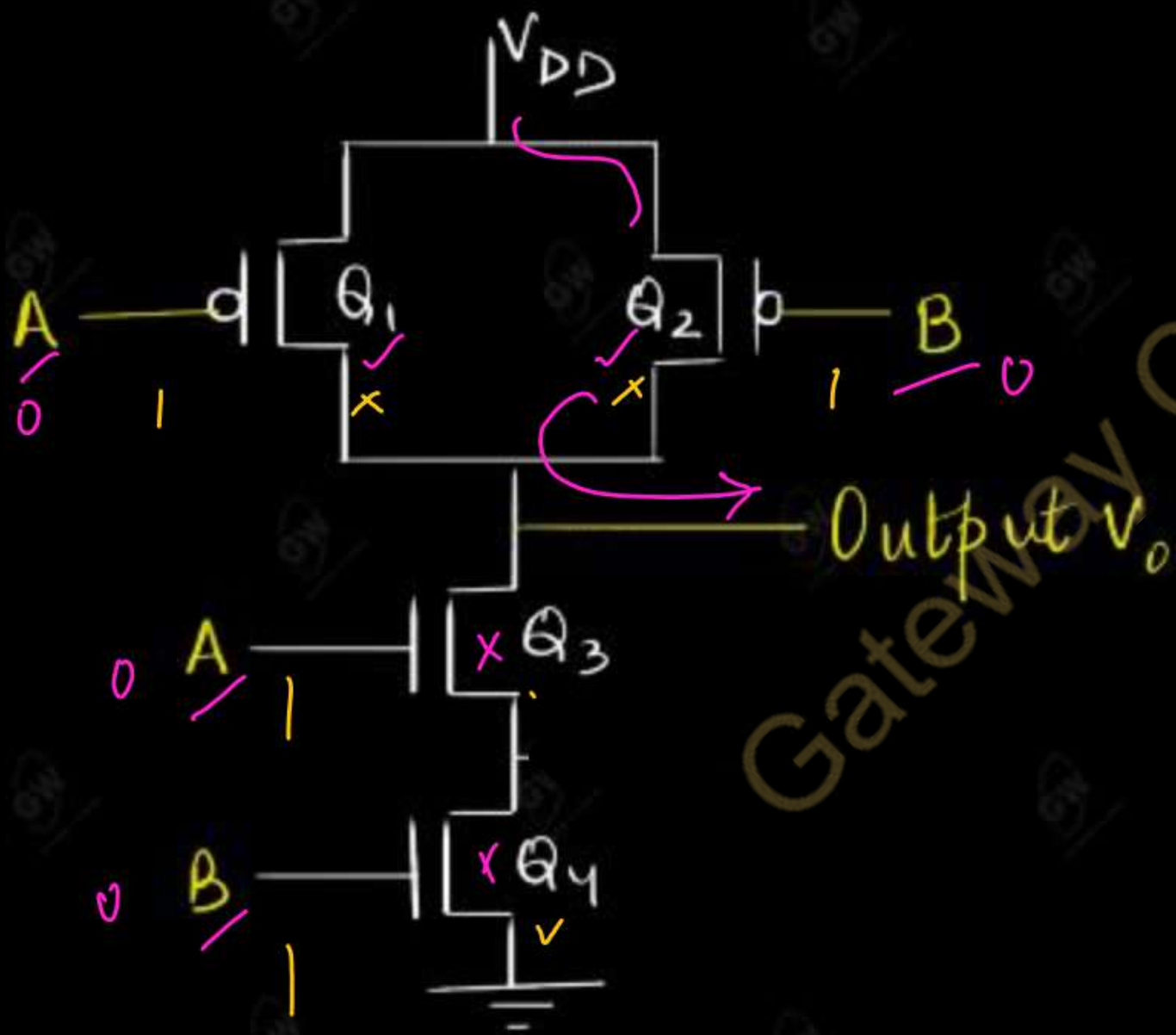


V_{in}	V_o
0	1
1	0

CMOS NAND Gate → P N P S

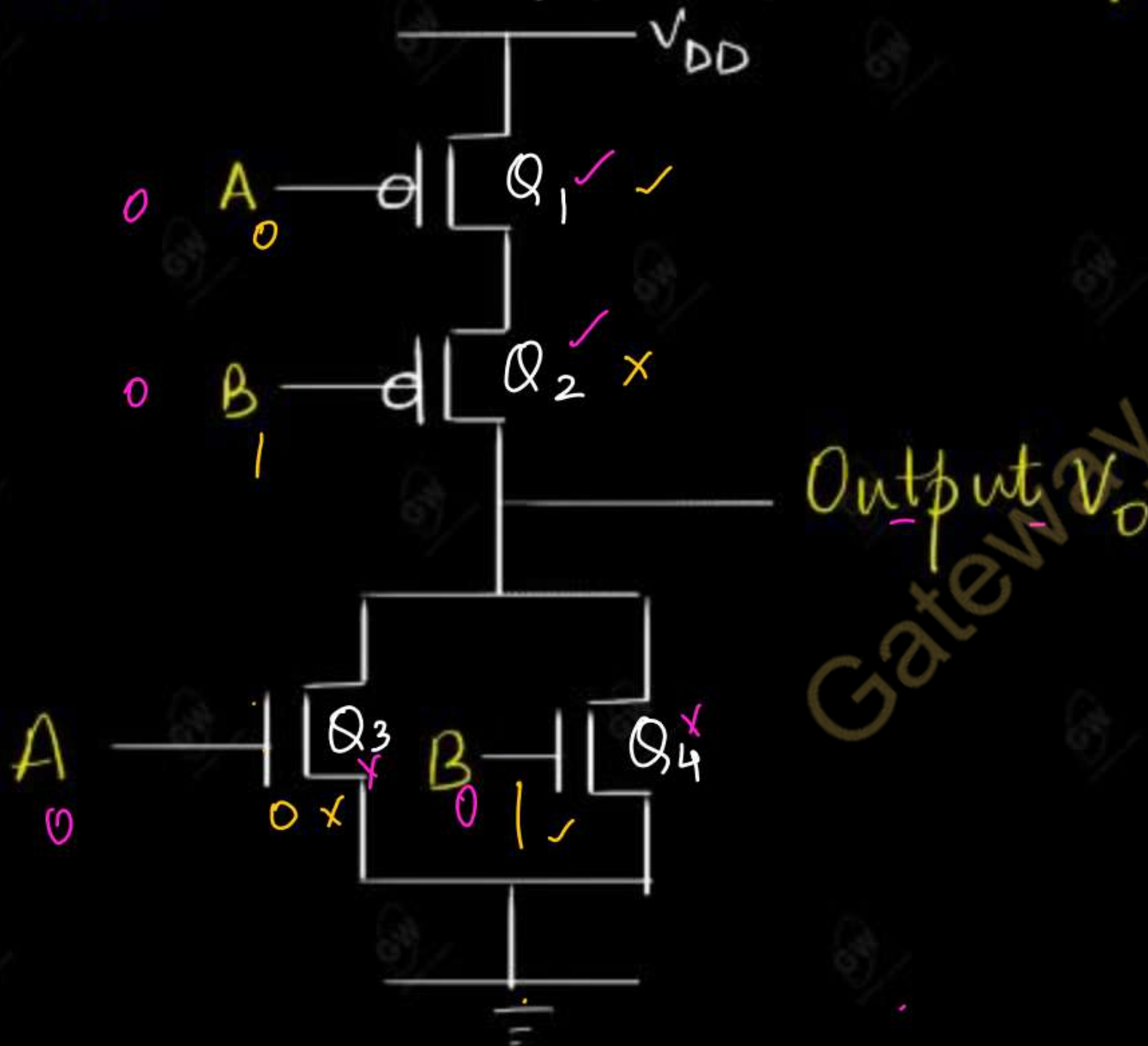
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It consists of **two p**-channel MOSFETs, Q1 and Q2, connected in **parallel** and **two n**-channel MOSFETs, Q3 and Q4 connected in **series**.



A	B	Q1	Q2	Q3	Q4	V _o
0	0	ON	ON	OFF	OFF	1
0	1	ON	OFF	OFF	ON	1
1	0	OFF	ON	ON	OFF	1
1	1	OFF	OFF	ON	ON	0

It consists of **two p-channel MOSFETs**, Q1 and Q2, connected in **series** and **two n-channel MOSFETs**, Q3 and Q4 connected in **parallel**.



A	B	Q1	Q2	Q3	Q4	V _o
0	0	ON	ON	OFF	OFF	1
0	1	ON	OFF	OFF	ON	0
1	0	OFF	ON	ON	OFF	0
1	1	OFF	OFF	ON	ON	0

Que – Compare TTL and CMOS in terms of Power Consumption and Speed.

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CMOS circuits have extremely **low power consumption**, because only one transistor in each pair conducts at a time.

TTL (Higher Power): TTL circuits consume higher power compared to CMOS. They are less energy-efficient and generate more heat than CMOS.

CMOS has **slower speed** as compared to TTL. TTL has faster switching speeds due to the direct, saturated coupling of Bipolar Junction Transistors (BJTs).

Que – Why ECL is better than other logic family ?

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ECL is better than other logic family because of its superior speed (**fastest logic family**) due to its non-saturating Transistors.

In other words the transistors in ECL are prevented from going into saturation and always work in cut-off mode and active mode, Thus reducing the propagation delay to less than 1 nsec thereby increasing its speed.

Memory

Memory is a device that is used to store data and instructions in the digital systems like computers and other microprocessor-based systems.

Types of Memory



ROM : Read Only Memory: Its non volatile memory, i.e, the information stored in it, is not lost even if the power supply goes off. It's used for the permanent storage of information.

Types of ROMs :

- (i) PROM :** It's Programmable ROM. Its contents are decided by the user. The user can store permanent programs, data etc in a PROM.
- (ii) EPROM :** An EPROM is an erasable PROM. The stored data in EPROM's can be erased by exposing it to UV light for about 20 min. It's not easy to erase it because the EPROM IC has to be removed from the computer and exposed to UV light.
- (iii) EEPROM (Electrically Erasable PROM) :** The chip can be erased & reprogrammed on the board easily byte by byte. It can be erased within a few milliseconds.

The internal structure of a Read-Only Memory (ROM) used to implement a 3-variable Boolean function consists primarily of an **Address Decoder** (which acts as a fixed AND array) and a **Programmable OR gate array**.

For a 3-variable function (A, B, C), the ROM is designed as an $8 \times n$ ROM, where 8 (2^3) represents the number of minterms, and n is the number of output bits.

Example :- Implement

$$F = AB + A'BC'$$

$$G = A'B'C + C'$$

$$H = AB'C' + ABC' + A'B'C$$

Using ROM.

$$F = AB + A'BC'$$

$$G = A'B'C + C'$$

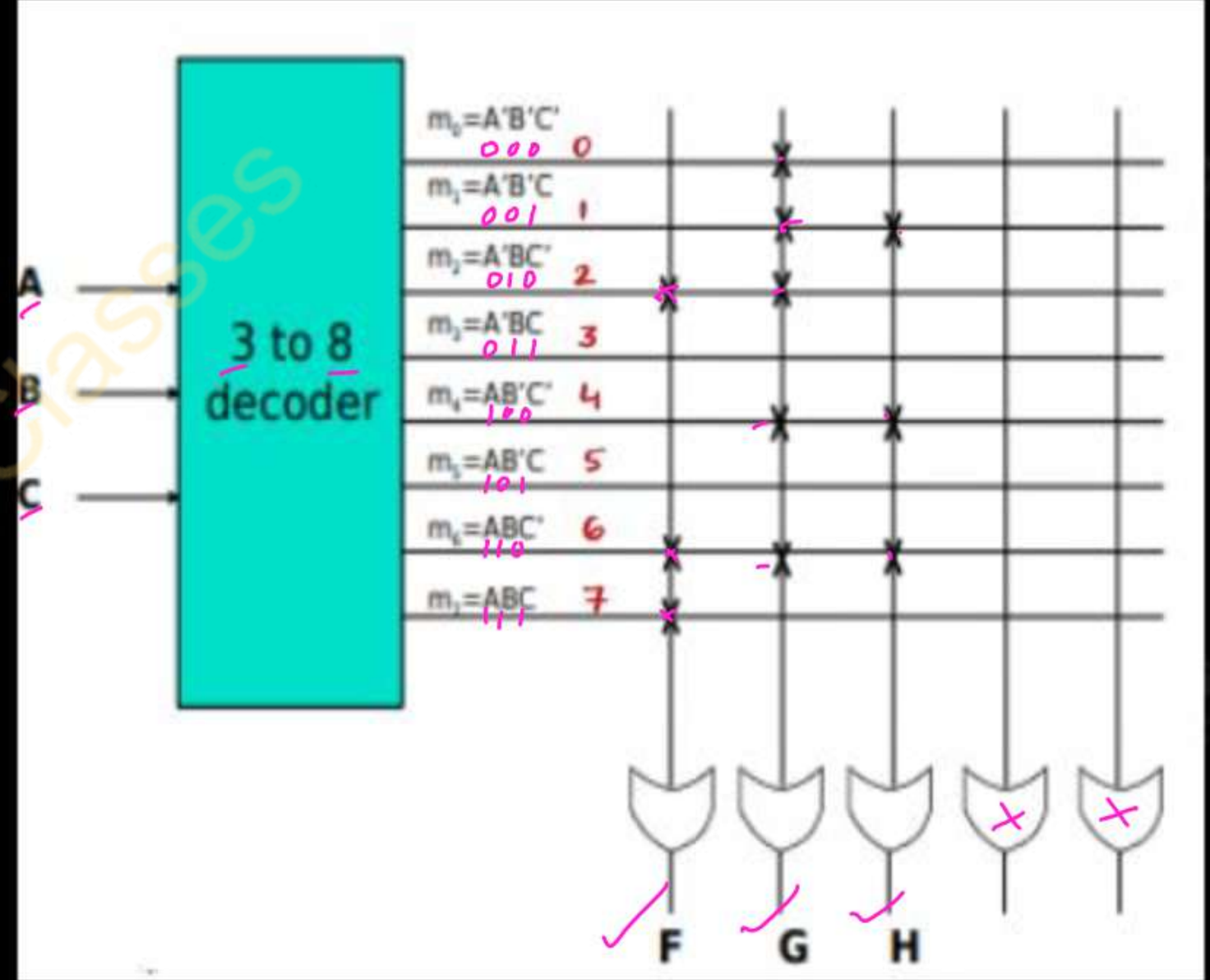
$$H = AB'C' + ABC' + A'B'C$$

$ABC + A'BC'$
 $ABC' + A'BC'$

3 inputs

3 outputs

A	B	C	F	G	H
0	0	0	0	1	0
0	0	1	0	1	1
0	1	0	1 ✓	1	0
0	1	1	0	0	0
1	0	0	0	1	1
1	0	1	0	0	0
1	1	0	1 ✓	1	1
1	1	1	1	0	0



EEPROM (Electrically Erasable Programmable Read-Only Memory) is a non-volatile memory that stores data even without power, allowing individual bytes to be erased and rewritten electrically, making it perfect for storing settings and small user data in devices like microcontrollers, smart cards, and remote controls.

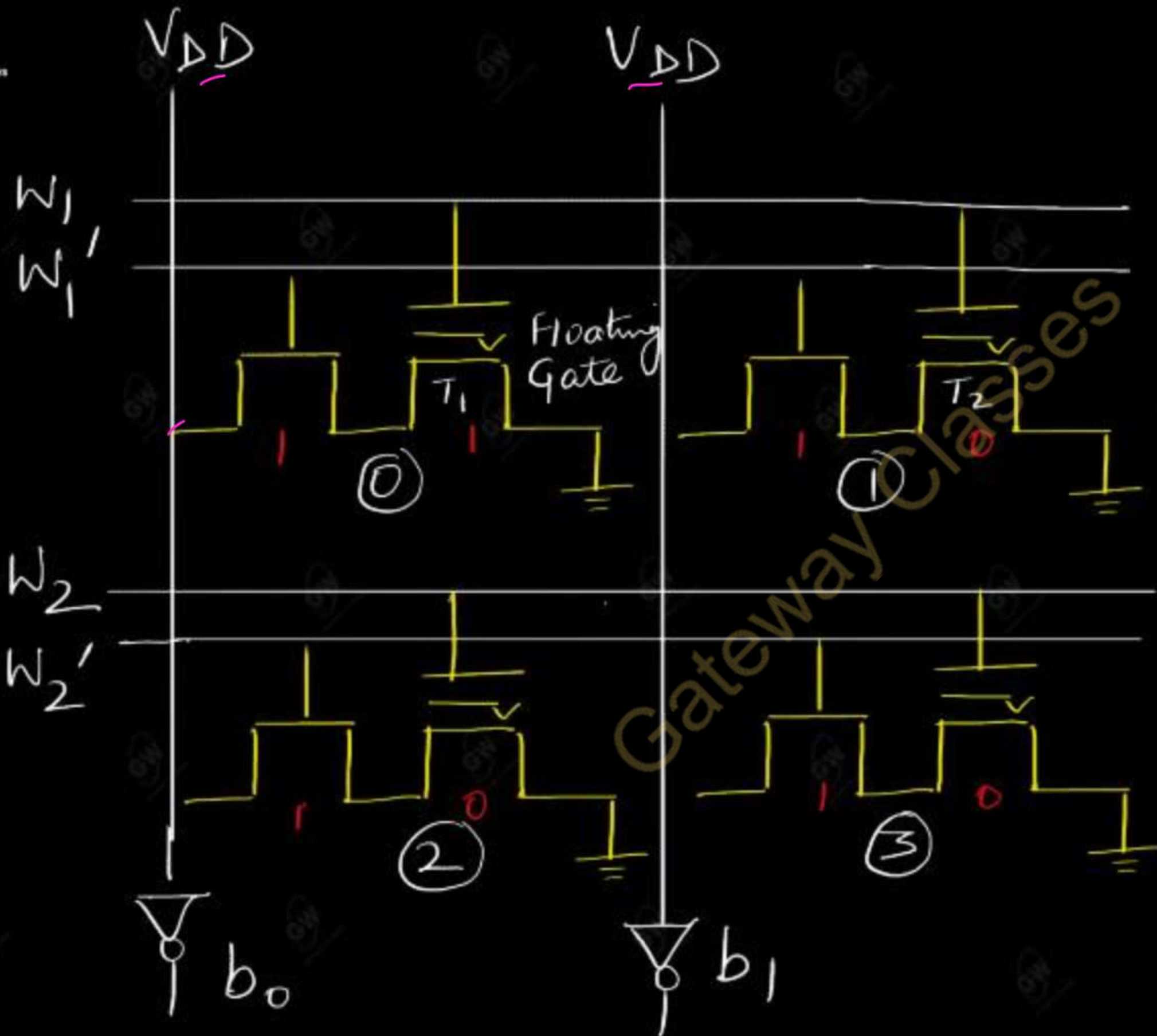
Unlike older EPROMs needing UV light, EEPROMs use electrical signals for reprogramming.

Common Uses

Storage device

Microcontrollers and embedded systems.

Smart cards, remote controls, and battery-powered devices



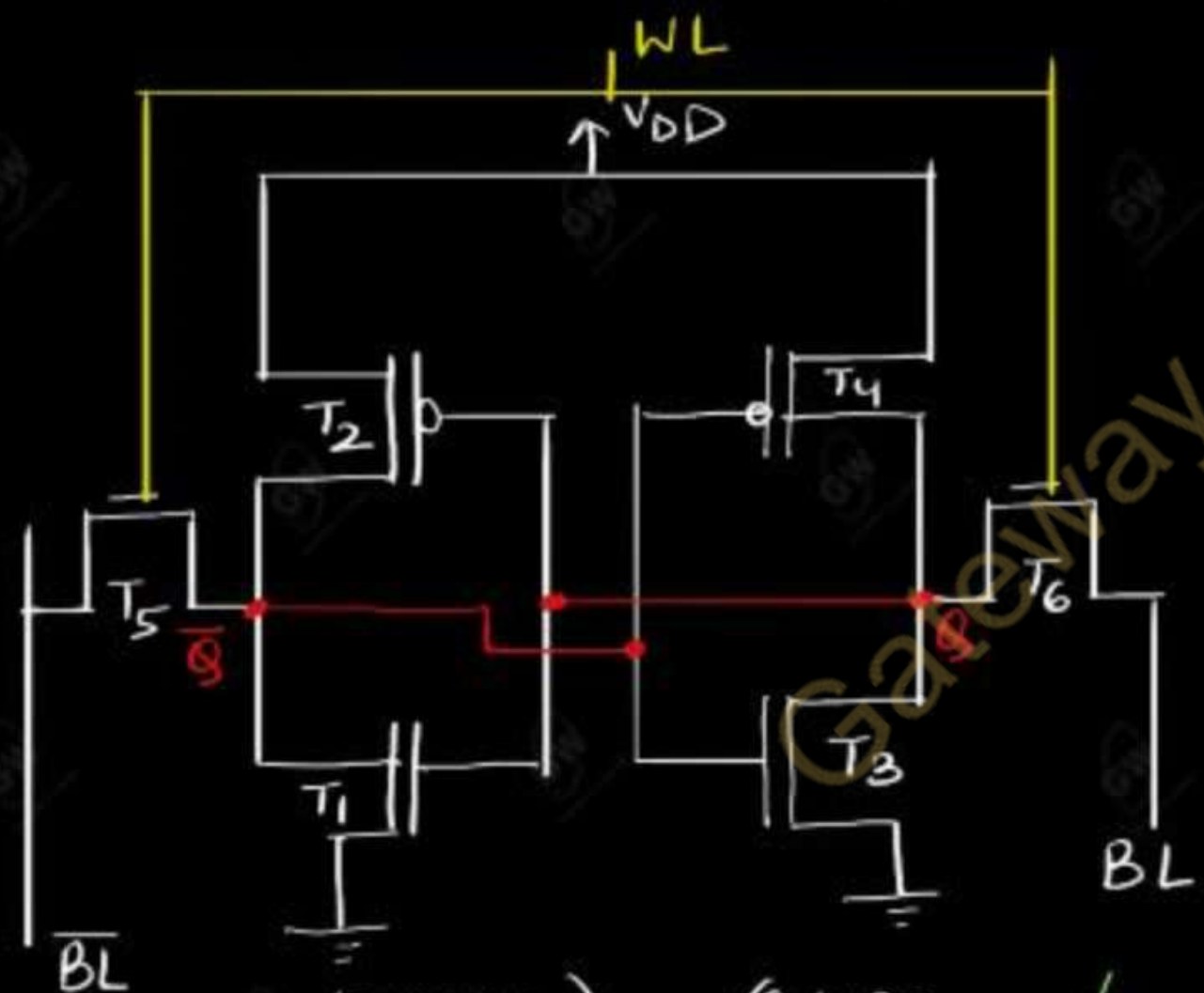
say
Already programmed
to 11, 10, 10, 10
 $V_{DD} \rightarrow$ select line
for cells (0-4)
To erase

SRAM

volatile

(Data remains stored till there is power)

Data is stored in static Latches (Transistors)



(Data remains stored till there is power)

CMOS1
CMOS2
Volatile

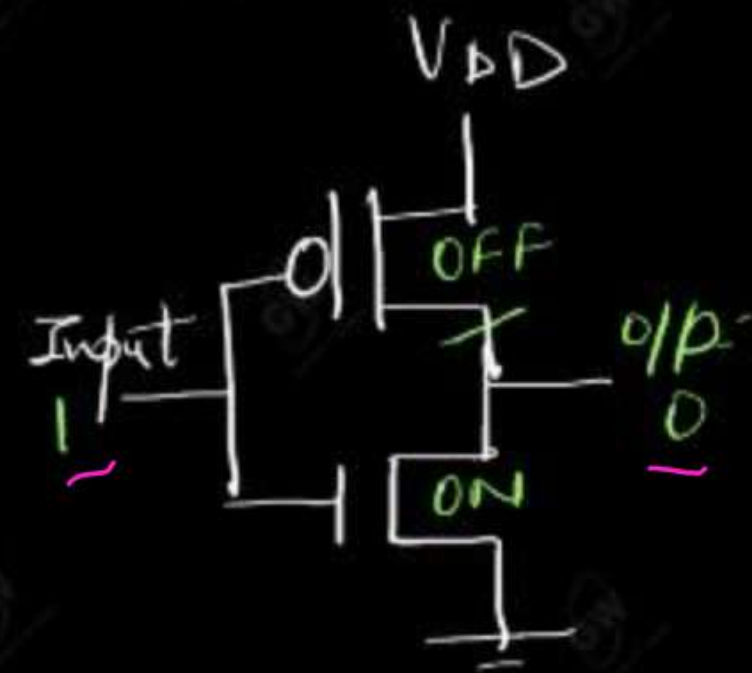
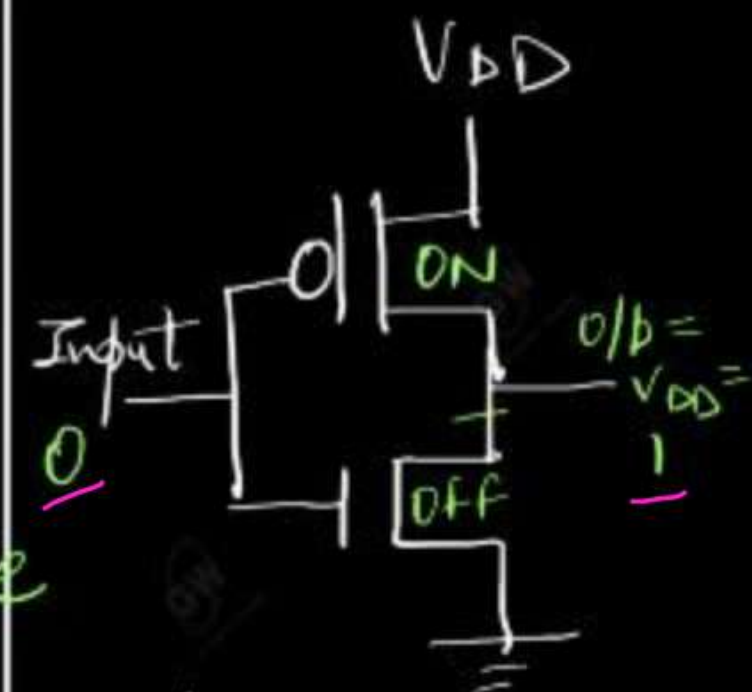
T_5 and T_6 are access transistors

WL $\rightarrow$ Word line
BL $\rightarrow$ Bit line

BL and $\overline{BL}$ are used for Read and write operation

BL and $\overline{BL}$ work as Input line for write operation

SRAM $\left\{ \begin{array}{l} \text{Hold} \\ \text{Read} \\ \text{Write} \end{array} \right.$



Static Random Access Memory

Read Operation → BL work as output line

Consider 1 is stored in cell. (no new value entered)

So, $Q = 1 (V_{DD})$, $Q' = 0$ (low), T_5 OFF, T_6 OFF, Latch, Hold

Before operation, BL & BL' are precharged to an intermediate voltage ($V_{DD}/2$).

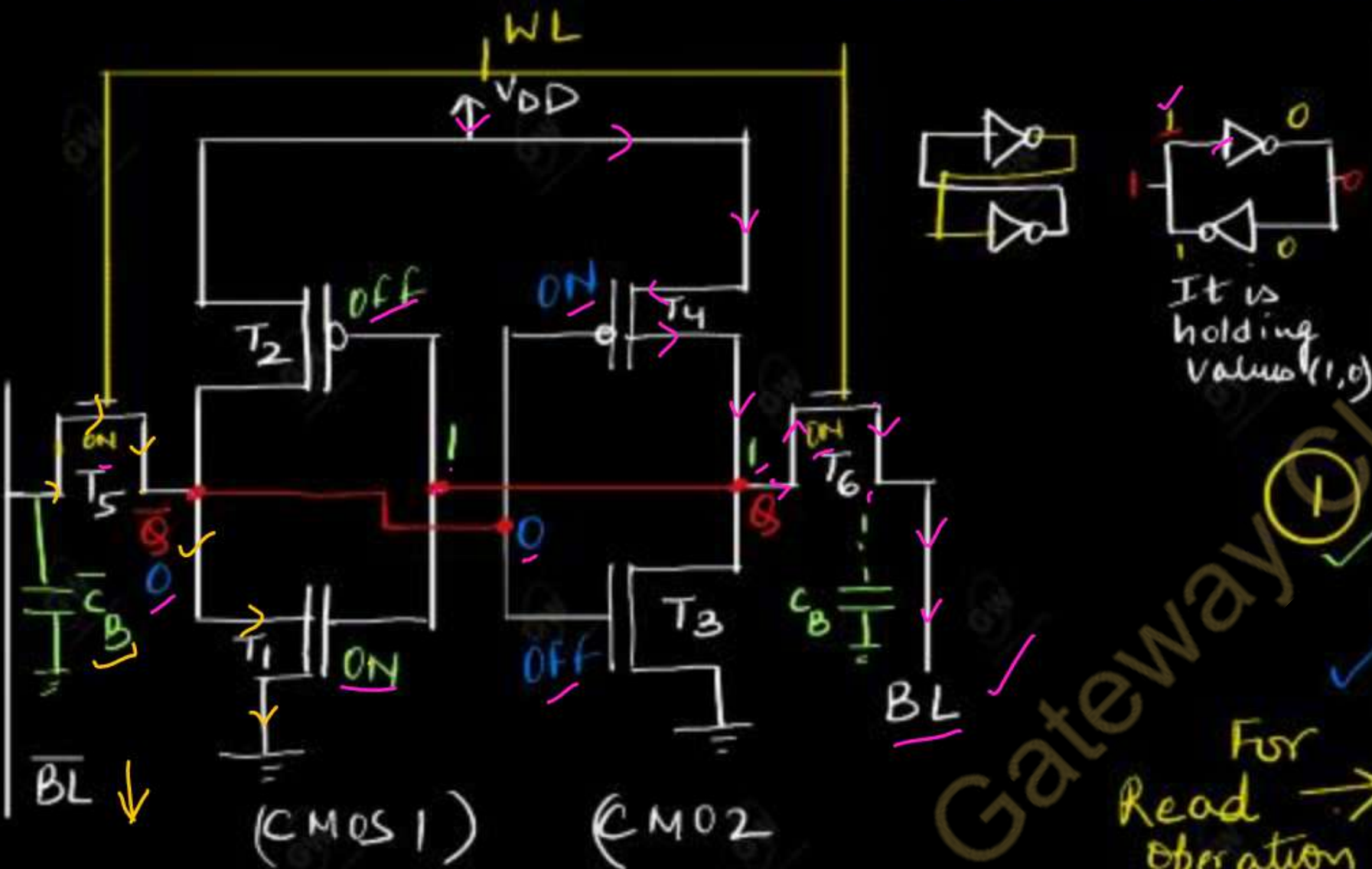
$Q=1$ So T_1 will be ON, $T_2 \rightarrow$ OFF

$Q'=0$ So $T_4 \rightarrow$ Turned ON, $T_3 \rightarrow$ OFF

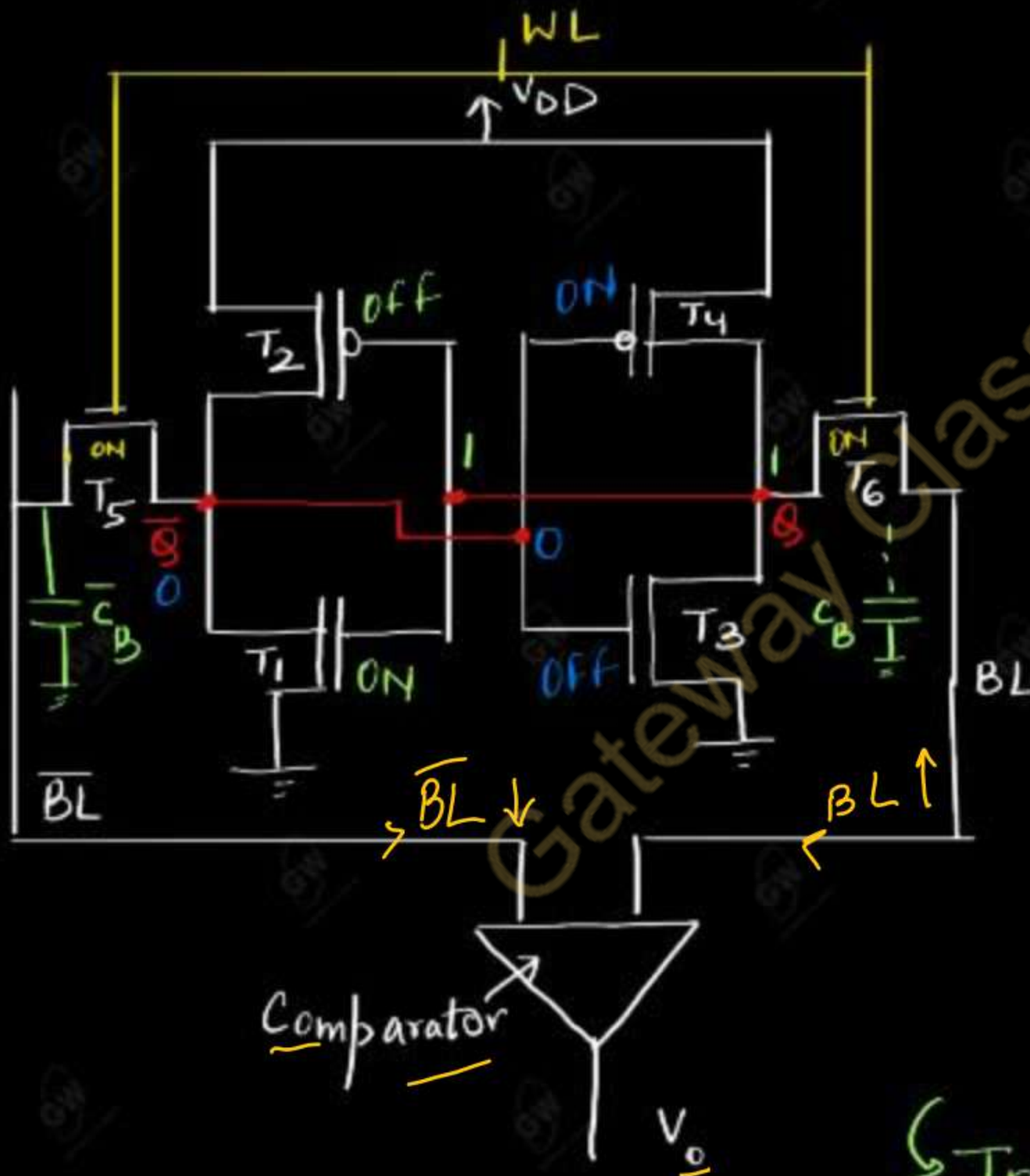
$WL = 1 (V_{DD}) \rightarrow T_5 \text{ \& } T_6$ will be ON

Current will flow from V_{DD} through T_4 & T_6 onto BL & charge the C_B .

Voltage across C_B will rise. BL will be more value at



② $\bar{Q} = 0$, hence current flows from $\bar{BL} \rightarrow T_5 \rightarrow T_1 \rightarrow$ Ground. $\bar{C}_B$ starts discharging. value at $\bar{BL}$ will be less



If $BL > \overline{BL}$
then $V_o = 1$

i.e. we read 1

and vice-versa

→ To read 0, $Q = 0$, $\overline{Q} = 1$, LHS of ckt
 $\overline{BL} > BL$, $V_o = 0$, we read 0

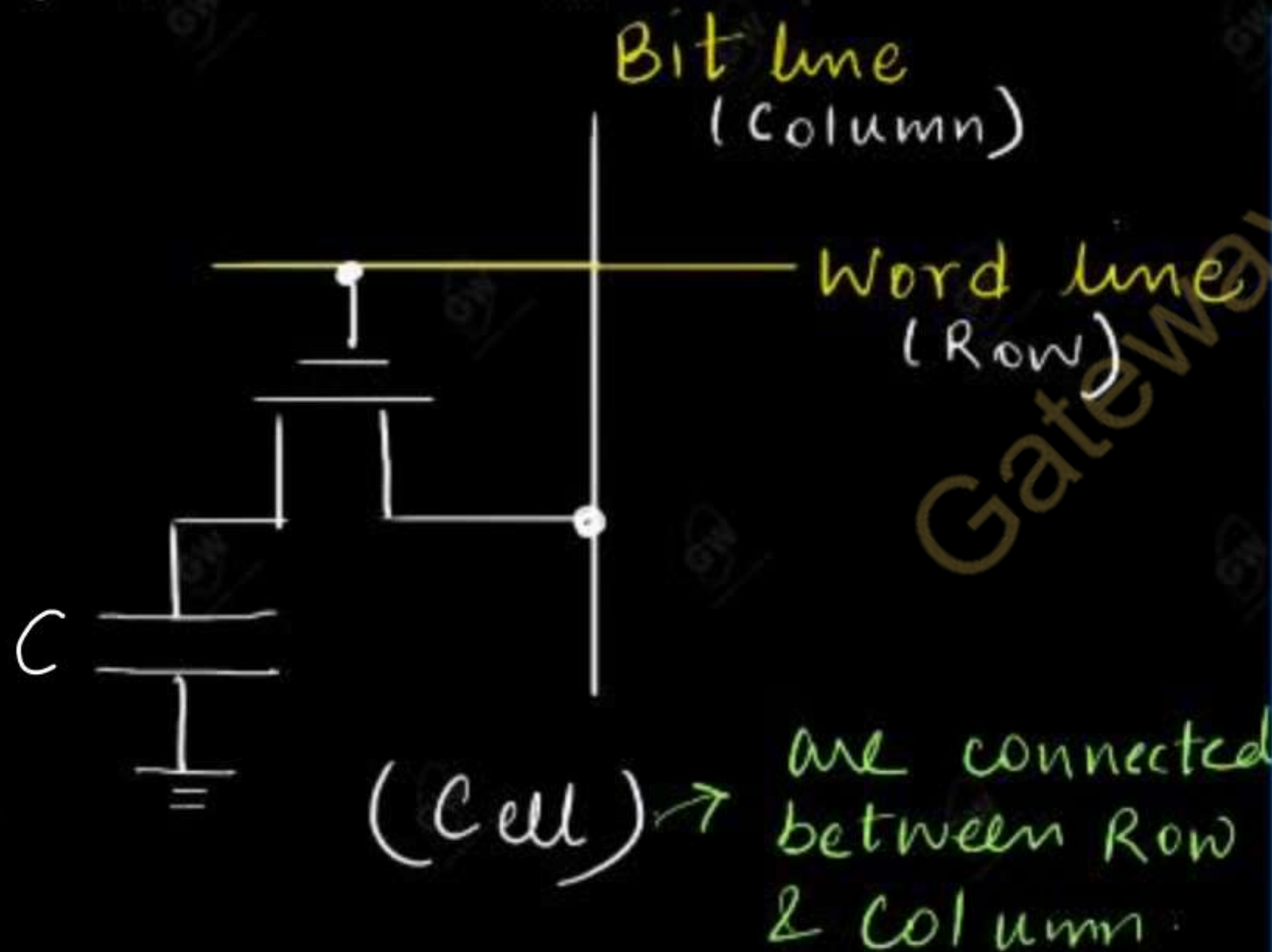
DRAM

Dynamic Random Access Memory

(Data remains stored till there is power)
If power is off, data is removed.

← Volatile

→ Data is stored as charges in Capacitors (Needs to refresh again & again)



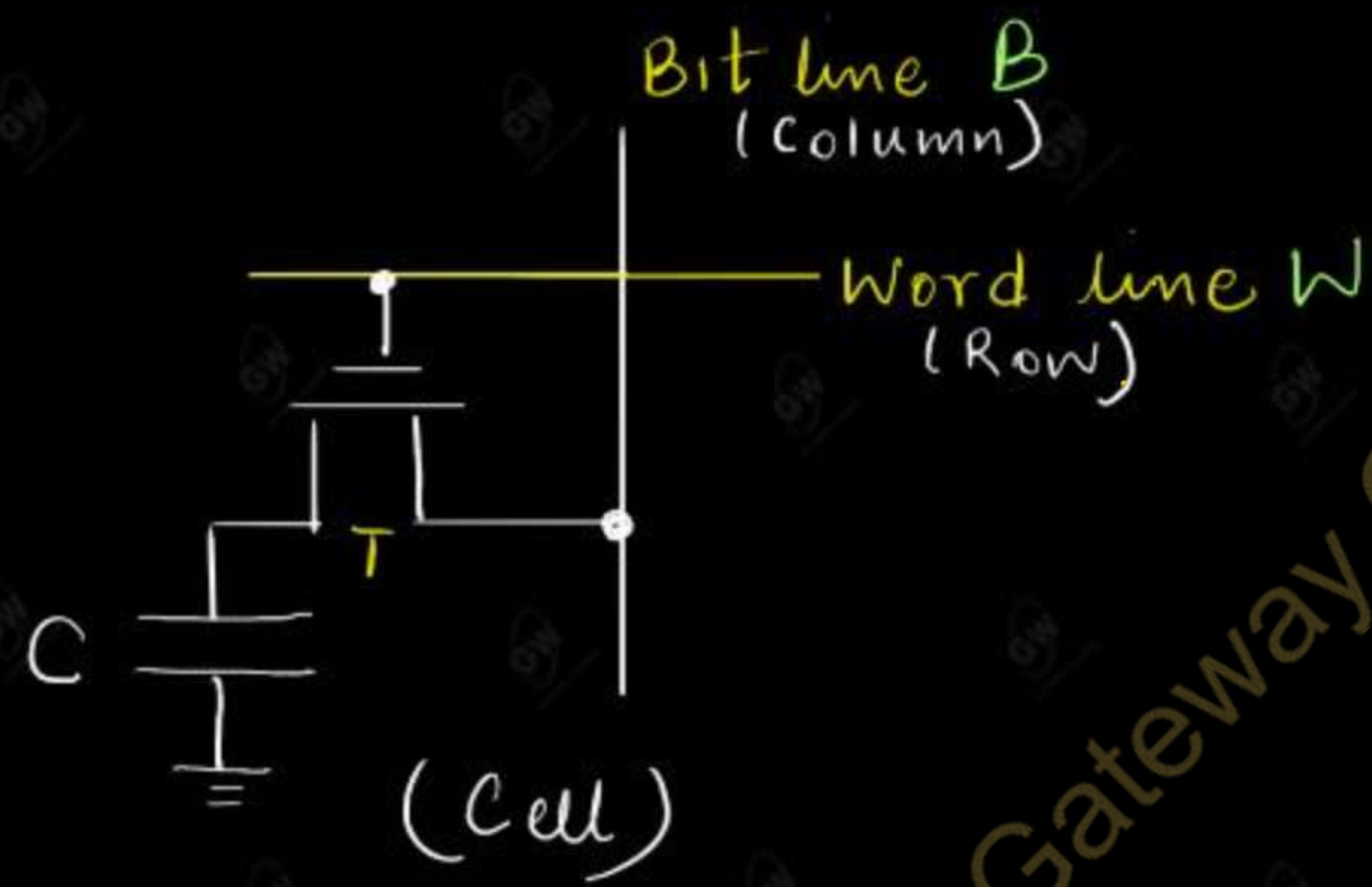
DRAM { SRAM
1 MOS, 1C } 6 Transistors

Area of cell is less in DRAM

means

sameⁱⁿ area, no. of cells of DRAM are more as compared to SRAM

↓
∴ Density of DRAM is more

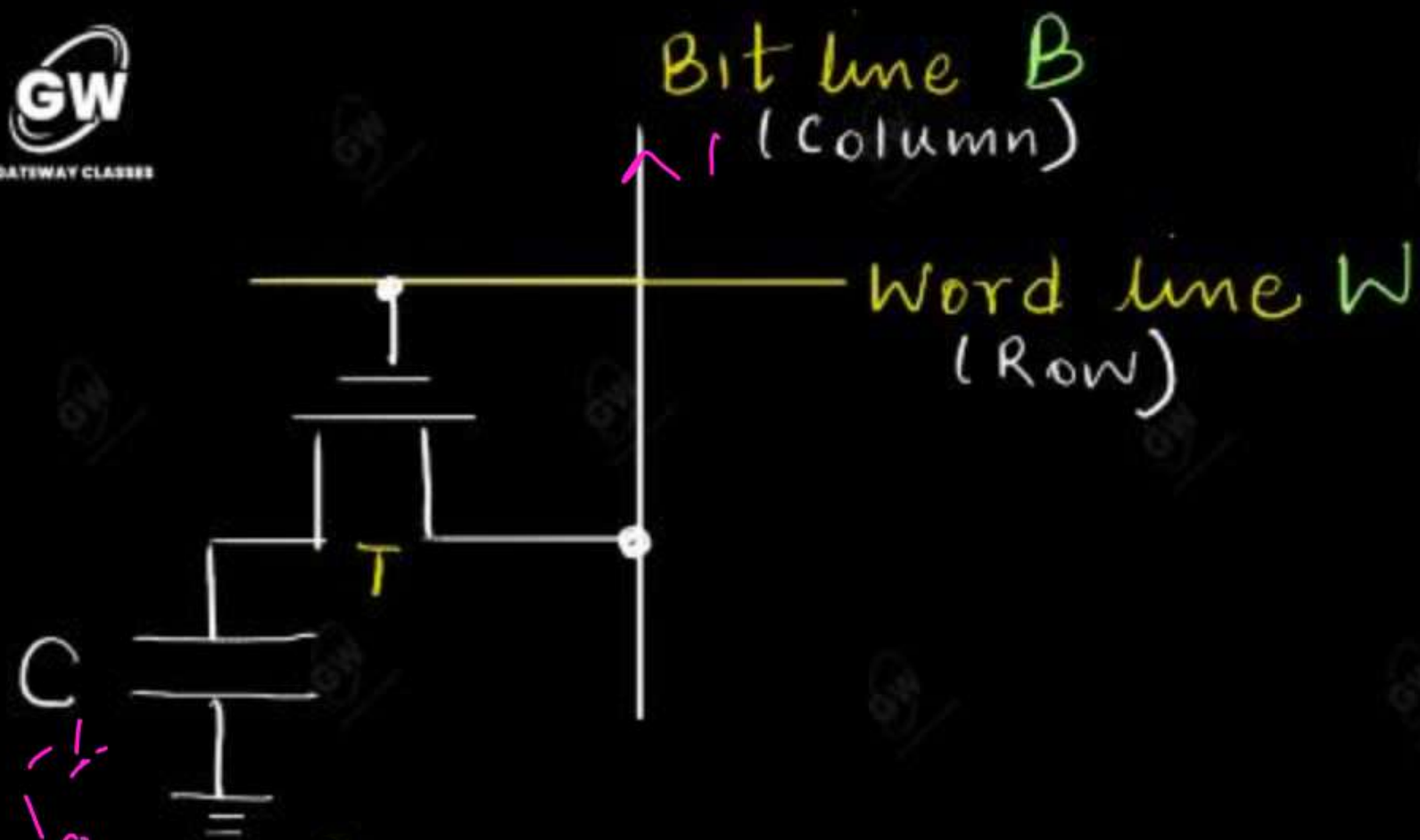


W should be 1 for
Read or Write operation

B works as

Input line for Write operation

Output line for Read operation



READ

W=1, T ON

B connects to C

If C is charged, $C=1$,

$B=1$

If C is discharged, $C=0$

$B=0$

HOLD

When $C=1$, we made

W=0 T OFF,

Row not connected, only column is connected, so circuit won't transfer data, it will hold data i.e. $C=1$.

WRITE W=1, T ON

say we want to write 0

Input line B=0 ✓
C discharges i.e. we wrote 0 on C

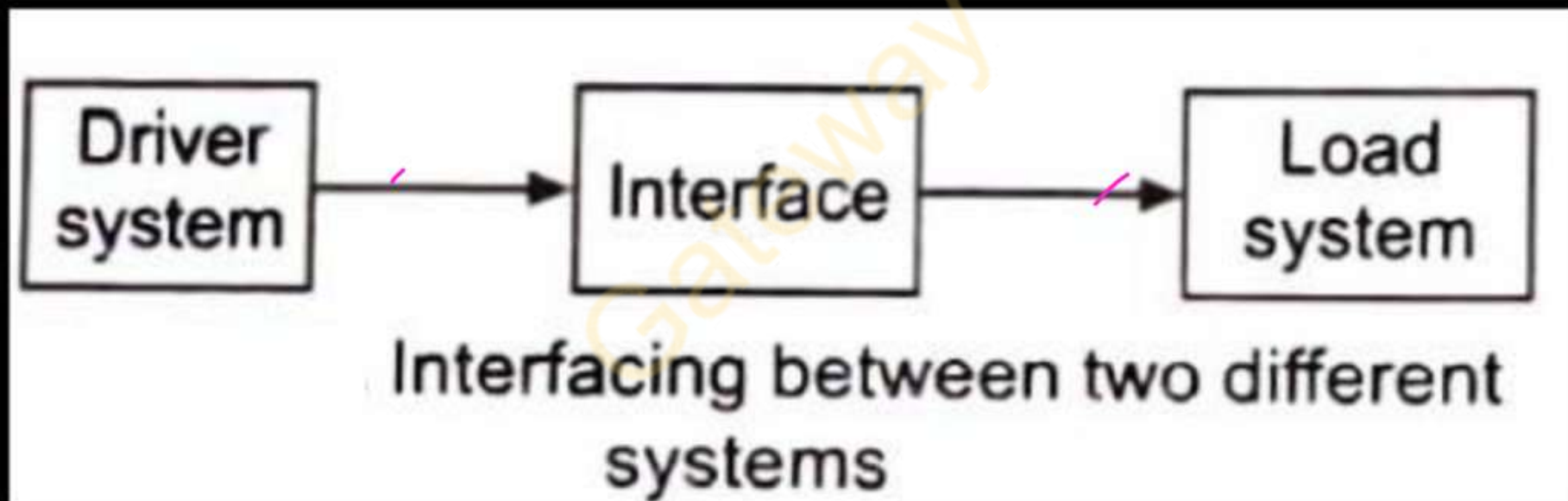
say we want to write 1

$B=1$, T was ON, ✓
C starts charging ∴ 1 is written on C

Parameters	SRAM	DRAM
Power Supply to Retain Data	Requires a consistent power supply	Needs electrical refresh periodically
Construction	A flip-flop circuit consisting of 6 transistors	A single capacitor and a transistor
Structure and Design Complexity	Complex	Simple
Power Consumption	Low	High
Storage Capacity	Lower storage capacity (1 MB to 16 MB)	High storage capacity (4 GB to 16 GB)
Speed/Access Time	Faster	Slower
Number of Transistors	6	1
Charge Leakage	No	Yes, if not refreshed regularly
Cost	Expensive	Less expensive
Application	Used in the L2 and L3 cache memory.	Used in a computer's primary memory

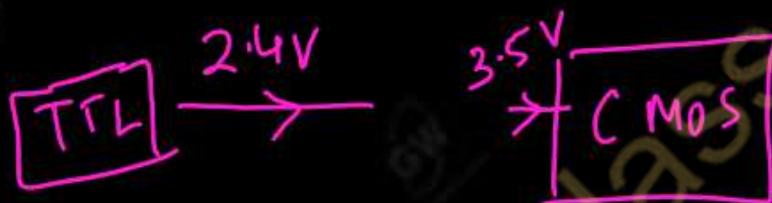
What is Interfacing?

- Interfacing = connecting the output of one logic family to the input of another.
- Needed when electrical characteristics differ (voltage levels, current levels).
- An interface circuit adjusts the signal so that the driver output becomes compatible with the load input.



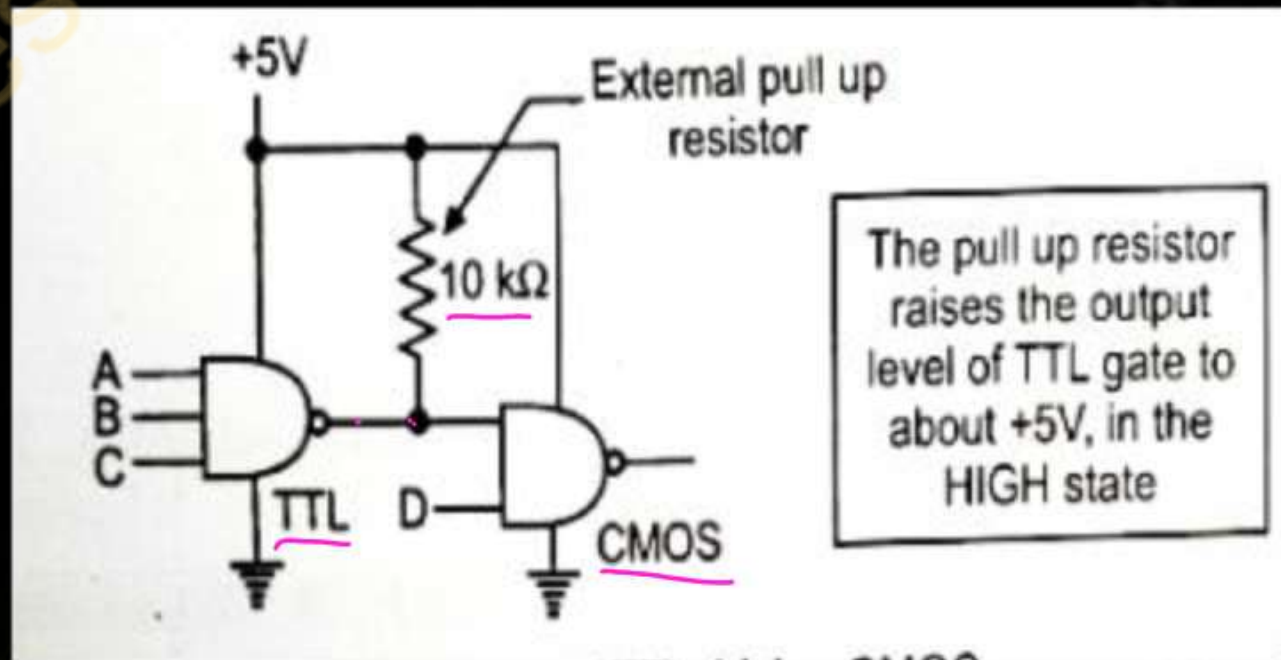
Basic Issue

- TTL and CMOS have different voltage levels.
- Current is not a problem because TTL output current > CMOS input current.
- **Voltage mismatch** is the major issue:
 - TTL HIGH output ≈ 2.4 V (min).
 - CMOS HIGH input requires ≥ 3.5 V.



Solution — Pull-up Resistor

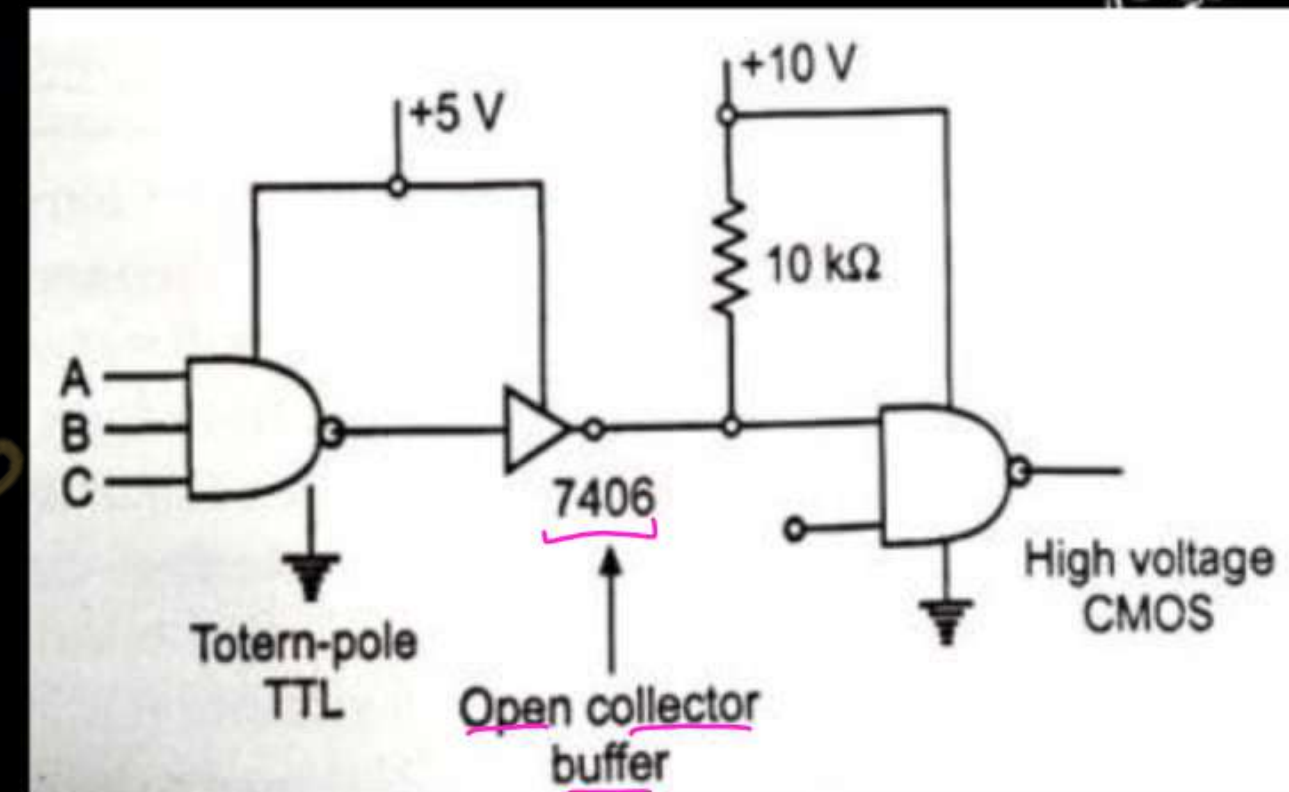
- A pull-up resistor (≈ 10 k Ω) is connected from TTL output to +5V.
- This raises TTL HIGH level to approx. 5 V $\rightarrow$ suitable for CMOS.



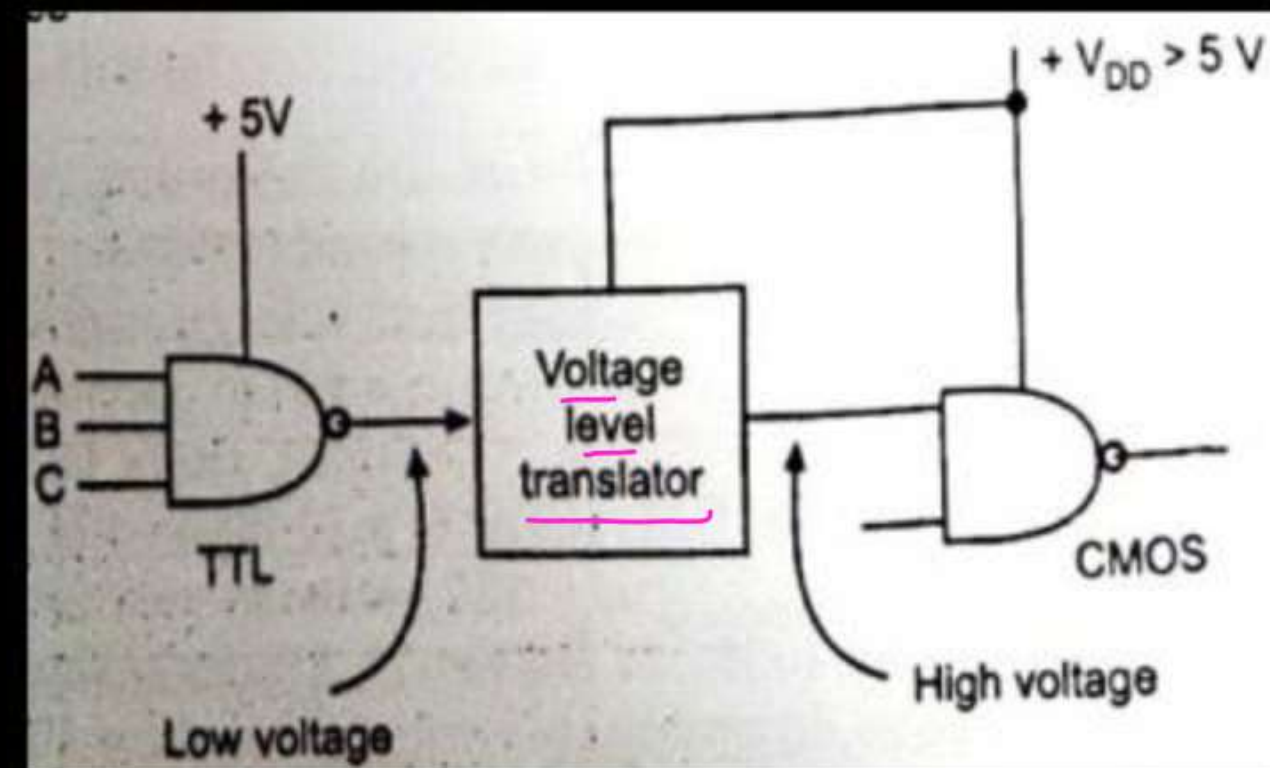
Problem:- High-voltage CMOS (e.g., $V_{DD} = 10\text{--}15$ V) requires HIGH input ≈ 7 V or more. TTL cannot output > 5 V, so pull-up method cannot be used.

Solution —② **Open Collector Buffer (7406)**

- Use 7406 (an inverting open-collector buffer) + pull-up resistor to high voltage.
- 7406 supports output up to 30V.
- Pull-up resistor raises the output to CMOS supply.

③ **Interfacing Using Level Shifters**

- A level shifter IC converts TTL 5V levels → CMOS higher voltage.
- Used when: Direct pull-up is not possible.



Problem

- CMOS output current ≈ 0.4 mA.
- TTL input current requires 1.6 mA.
- Thus, CMOS cannot sink enough current.

Solution

- Use a non-inverting buffer between CMOS and TTL.
- Buffer increases the current sinking capacity.

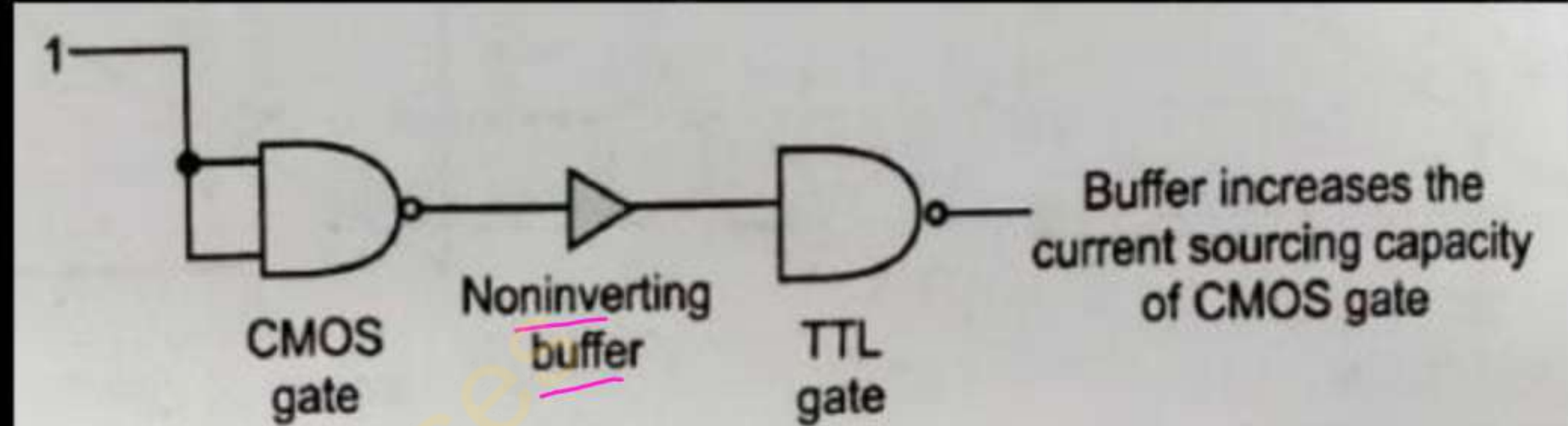


Fig. 16.79. CMOS driving TTL in low state

Programmable Logic Device (PLD)

It is an Integrated circuit (IC) that can be configured or "programmed" by the user to implement specific logic functions .

PLDs are essentially a combination of AND gates and OR gates .

The connections within the AND array and OR array can be programmed (made or broken) by the user according to the required function.

PLDs

PROM (Programmable Read Only Memory)

PAL (Programmable Array Logic)

PLA (Programmable Logic Array)



(a) Programmable read-only memory (PROM)

FP



(b) Programmable array logic (PAL)

PF



(c) Programmable logic array (PLA)

PP

PLA (Programmable Logic Array): Both the AND array and the OR array are programmable.

PAL (Programmable Array Logic): The AND array is programmable, but the OR array is fixed (hardwired)

Differences between PLA and PAL

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PLA

PLA stands for Programmable Logic Array

They have a programmable AND array and programming OR array.

The speed of PLA is lower than PAL.

It is expensive.

PAL

PAL stands for Programmable Array Logic.

They have programmable AND array and **fixed** OR array.

The speed of PAL is higher than PLA.

It is not that expensive.

Implementation Using PLDs

Que -Implement the following

$$F1 = \sum m (0,3,4,7)$$

$$F2 = \sum m (1,2,5,7)$$

using

✓ (i) PLA

✓ (ii) PAL

✓ (iii) PROM

0-7 3 variable
K-Map

K Map for F1

AB \ C	00	01	11	10
0	1			1
1		1	1	

BC

$$F_1 = \bar{B}\bar{C} + BC$$

XNOR

K Map for F2

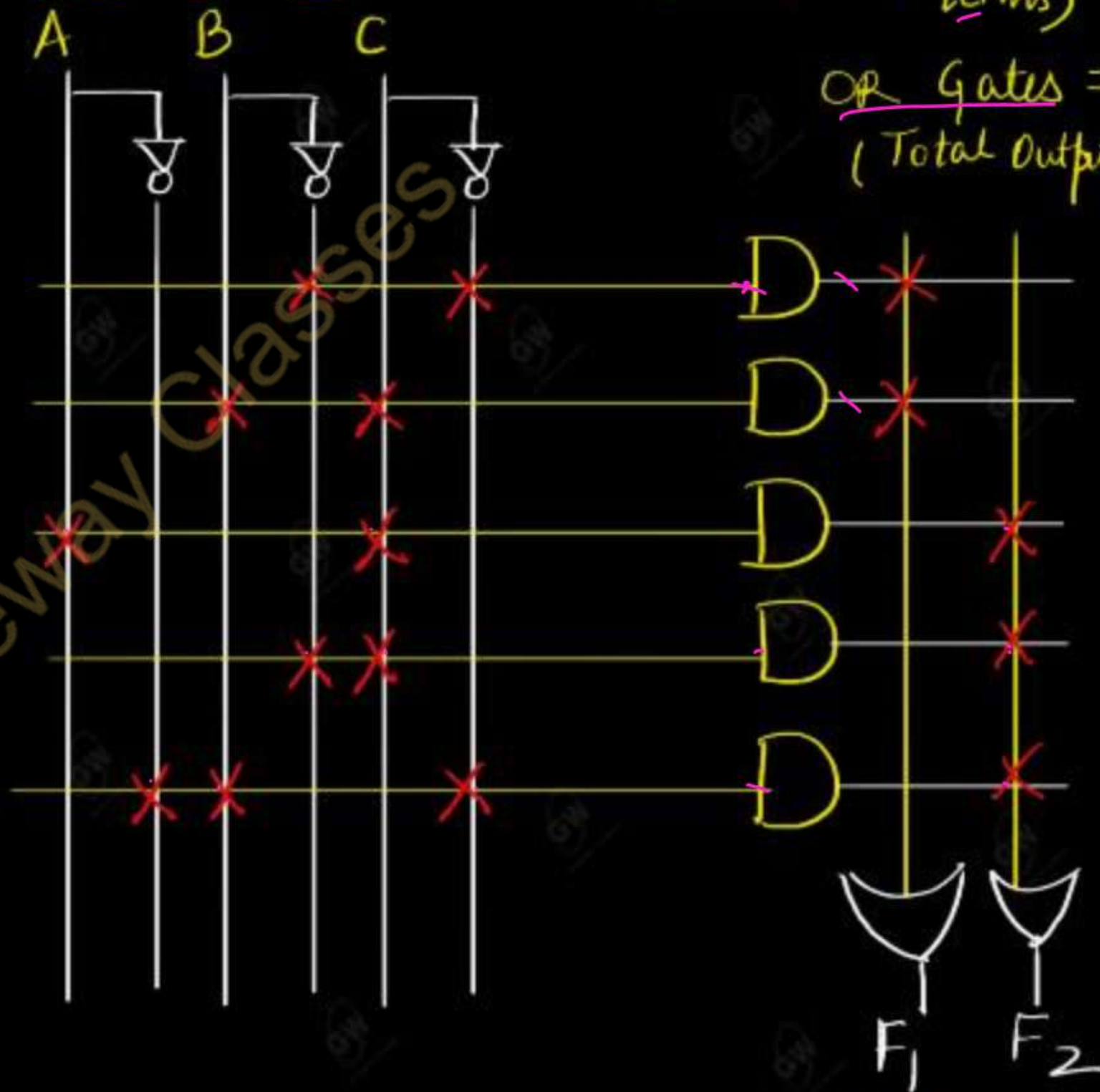
AB \ C	00	01	11	10
0		1		
1	1		1	1

$$F_2 = AC + \bar{B}C + \bar{A}B\bar{C}$$

Product Table

ALL PRODUCTS	INPUTS			OUTPUTS	
	A	B	C	F1	F2
$\bar{B}\bar{C}$	-	0	0	1	-
BC	-	1	1	1	-
AC	1	-	1	-	1
$\bar{B}C$	-	0	1	-	1
$\bar{A}B\bar{C}$	0	1	0	-	1

Implementation Using PLA



AND Gates = 5
(Total product terms)

OR Gates = 2
(Total outputs)

Product Table

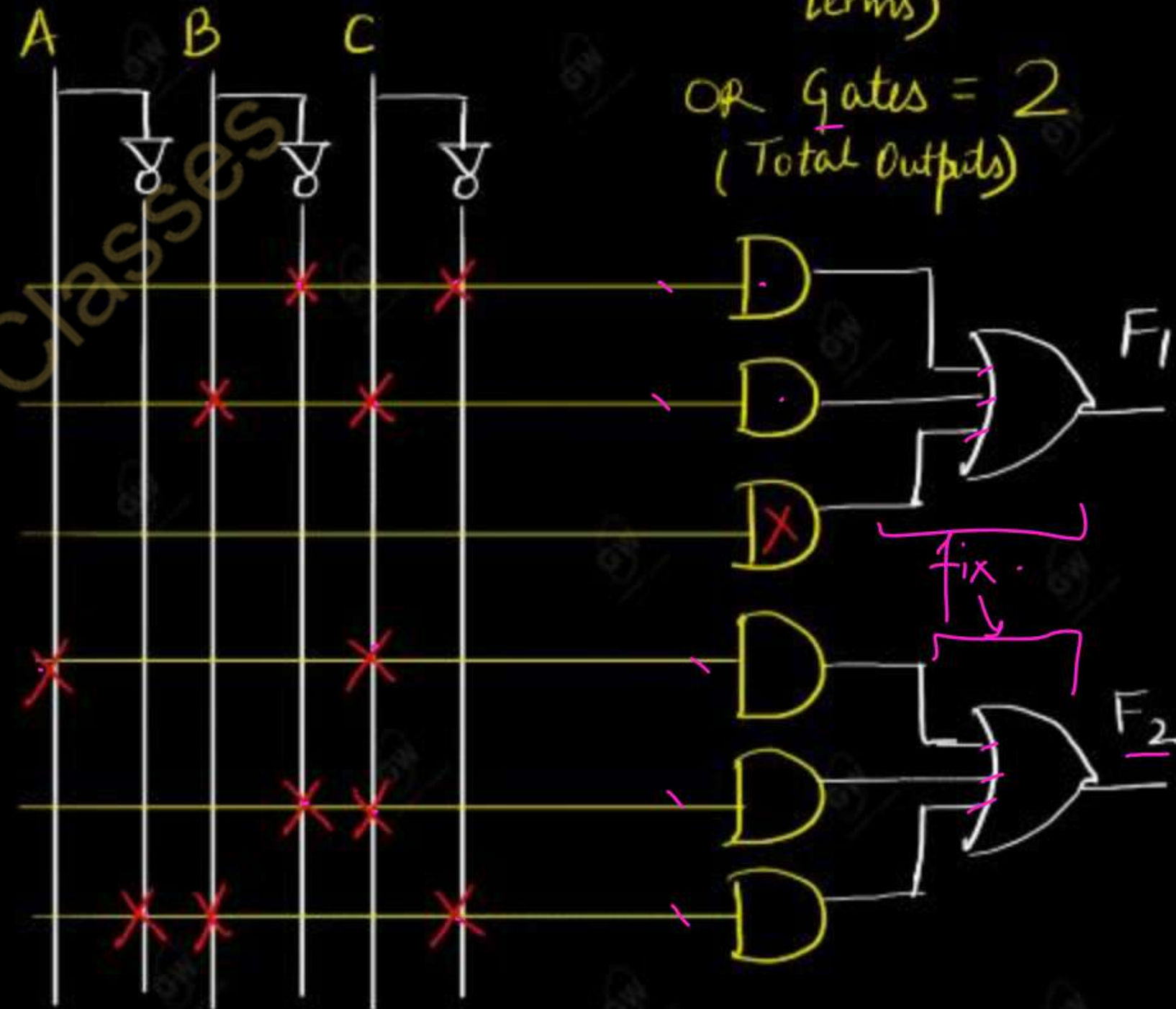
PAL

ALL PRODUCTS	INPUTS			OUTPUTS
	A	B	C	
$\bar{B}\bar{C}$	—	0	0	F1
BC	—	1	1	
AC	1	—	1	F2
$\bar{B}C$	—	0	1	
$\bar{A}B\bar{C}$	0	1	0	

Implementation Using PAL

AND Gates = 5
(Total product terms)

OR Gates = 2
(Total outputs)

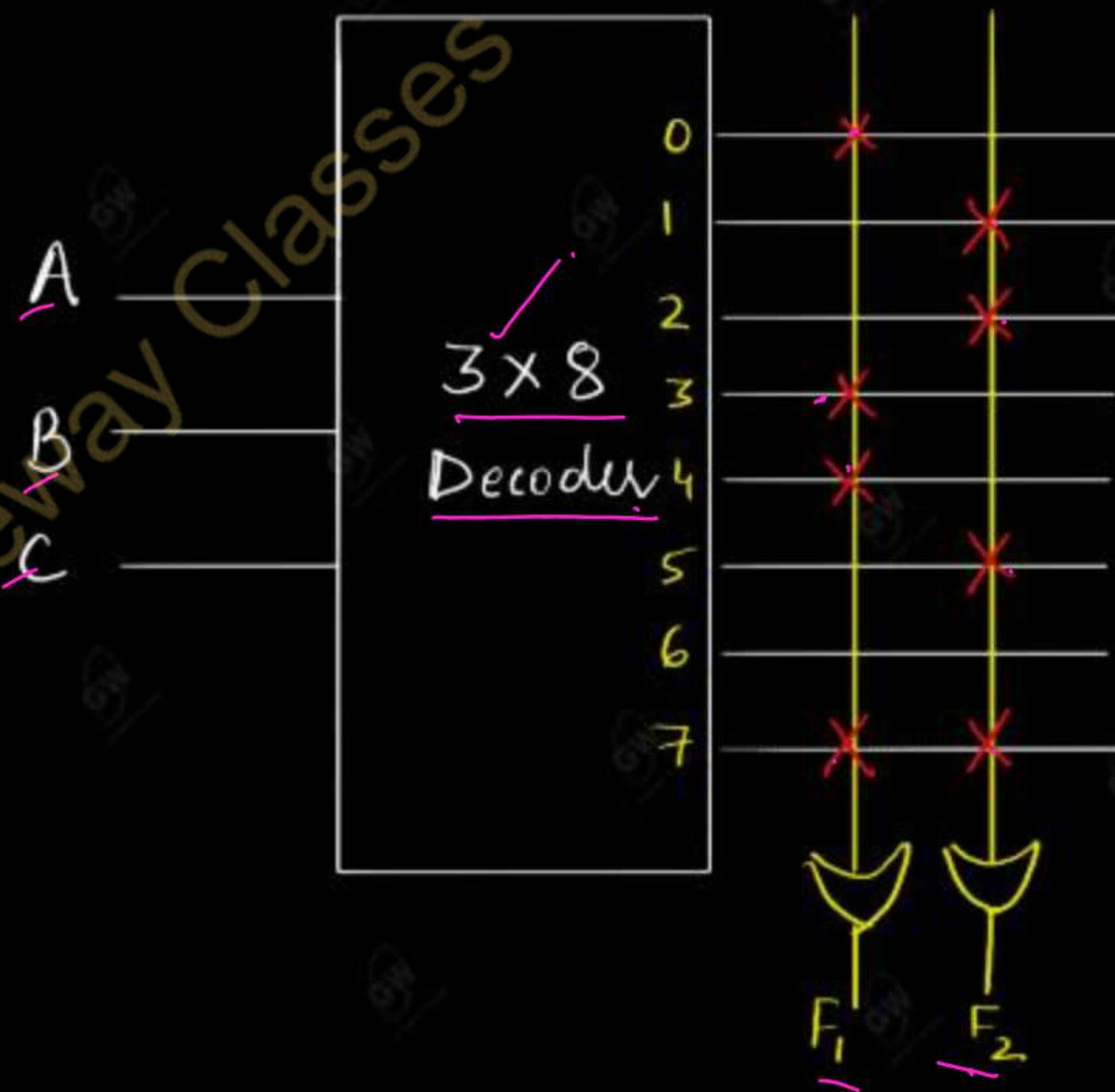


Implementation Using PROM

Given

$$F_1 = \sum m (0, 3, 4, 7)$$

$$F_2 = \sum m (1, 2, 5, 7)$$



Que – Implement a 3 bit Binary to Gray Code Converter using PLA .

Concept

$B \rightarrow G$

$B_3 \ B_2 \ B_1 \ B_0 \rightarrow G_3 \ G_2 \ G_1 \ G_0$

$$G_3 = B_3$$

$$G_2 = B_3 \text{ EX-OR } B_2 =$$

$$G_1 = B_2 \text{ EX-OR } B_1 =$$

$$G_0 = B_1 \text{ EX-OR } B_0 =$$

Similar bits 0
disimilar bits 1

EXAMPLE :-

$B \rightarrow G$

$0110 \rightarrow ?$

$B_3 \ B_2 \ B_1 \ B_0 \rightarrow G_3 \ G_2 \ G_1 \ G_0$

$$G_3 = 0$$

$$G_2 = 0 \text{ EX-OR } 1 = 1$$

$$G_1 = 1 \text{ EX-OR } 1 = 0$$

$$G_0 = 1 \text{ EX-OR } 0 = 1$$

Ans - 0101

EXAMPLE :-

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$B \rightarrow G$

$1010 \rightarrow ?$

$B_3 \ B_2 \ B_1 \ B_0 \rightarrow G_3 \ G_2 \ G_1 \ G_0$

$$G_3 = 1$$

$$G_2 = 1 \text{ EX-OR } 0 = 1$$

$$G_1 = 0 \text{ EX-OR } 1 = 1$$

$$G_0 = 1 \text{ EX-OR } 0 = 1$$

Ans 1111

$B \rightarrow G$

Binary Input			Gray Output		
B2	B1	B0	G2	G1	G0
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	1
0	1	1	0	1	0
1	0	0	1	1	0
1	0	1	1	1	1
1	1	0	1	0	1
1	1	1	1	0	0

K Map for G2

$B_2 B_1$				
	00	01	11	10
B_0 0			1	1
1			1	1

$$G_2 = B_2$$

K Map for G1

$B_2 B_1$				
	00	01	11	10
B_0 0		1		1
1		1		1

$$G_1 = \overline{B_2} B_1 + B_2 \overline{B_1}$$

K Map for G0

$B_2 B_1$				
	00	01	11	10
B_0 0		1	1	
1	1			1

$$G_0 = \overline{B_1} B_0 + B_1 \overline{B_0}$$

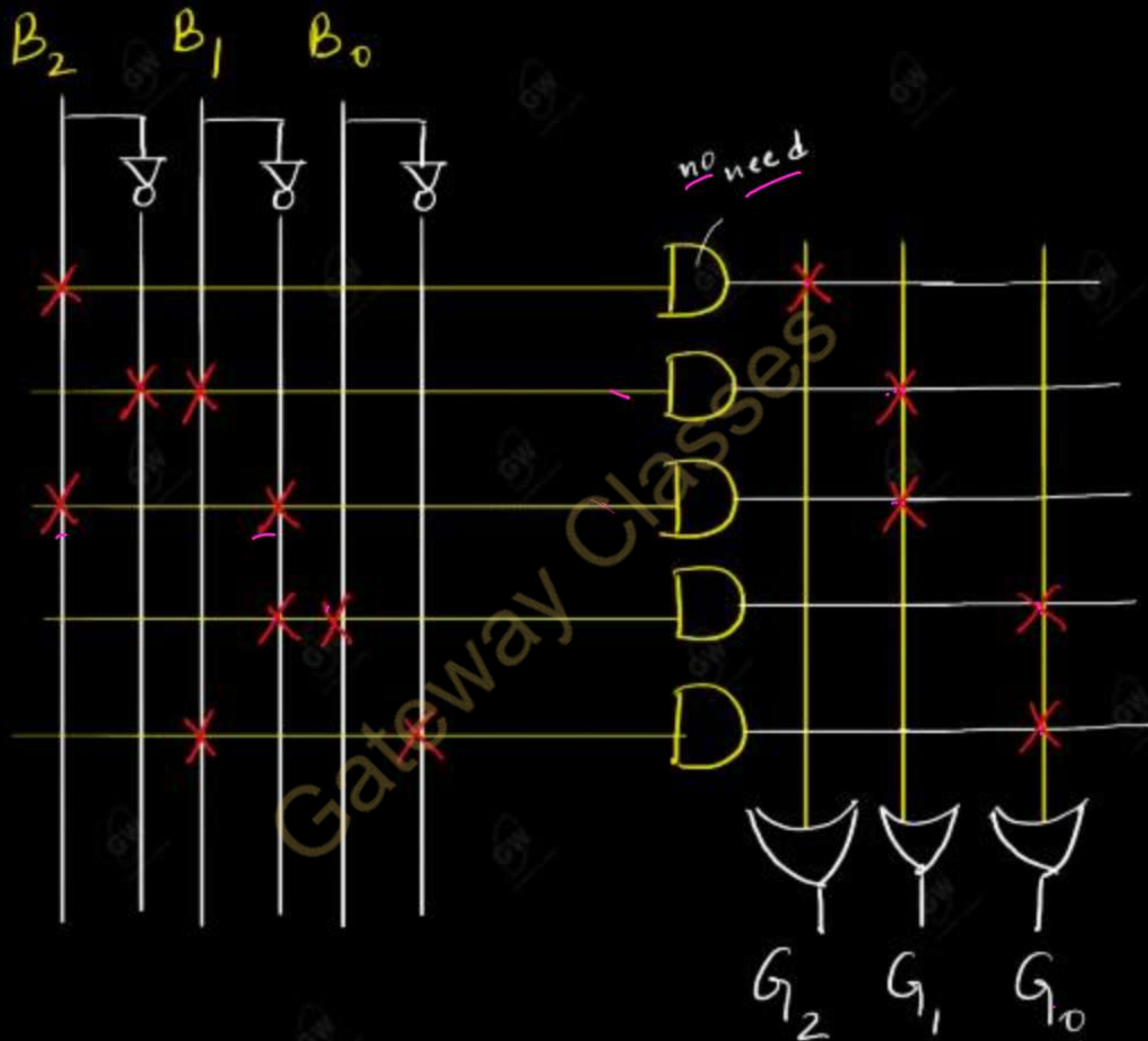
$$G_2 = B_2$$

$$G_1 = \bar{B}_2 B_1 + B_2 \bar{B}_1$$

$$G_0 = \bar{B}_1 B_0 + B_1 \bar{B}_0$$

AND Gates = $5 - 1 = 4$
(Total product terms)

OR Gates = 3
(Total Outputs)



$G \rightarrow B$

Que - Implement Gray Code to Binary Converter using PLA.

Concept

 $G \rightarrow B$ $G_3 \ G_2 \ G_1 \ G_0 \rightarrow B_3 \ B_2 \ B_1 \ B_0$ $B_3 = G_3$ $B_2 = B_3 \text{ EX-OR } G_2 =$ $B_1 = B_2 \text{ EX-OR } G_1 =$ $B_0 = B_1 \text{ EX-OR } G_0 =$

EXAMPLE :-

 $G \rightarrow GB$ $1101 \rightarrow ?$ $G_3 \ G_2 \ G_1 \ G_0 \rightarrow B_3 \ B_2 \ B_1 \ B_0$ $B_3 = 1$ $B_2 = 1 \text{ EX-OR } 1 = 0$ $B_1 = 0 \text{ EX-OR } 0 = 0$ $B_0 = 0 \text{ EX-OR } 1 = 1$

Ans **1001**

Gray Input			Binary Output		
G2	G1	G0	B2	B1	B0
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	1
0	1	1	0	1	0
1	0	0	1	1	1
1	0	1	1	1	0
1	1	0	1	0	0
1	1	1	1	0	1

K Map for B2

$G_2 G_1$	G_0			
	00	01	11	10
0			1	1
1			1	1

$$B_2 = G_2$$

K Map for B1

$G_2 G_1$	G_0			
	00	01	11	10
0		1		1
1		1		1

$$B_1 = \bar{G}_2 G_1 + G_2 \bar{G}_1$$

K Map for B0

$G_2 G_1$	G_0			
	00	01	11	10
0		1		1
1	1		1	

$$B_0 = \bar{G}_2 \bar{G}_1 G_0 + \bar{G}_2 G_1 \bar{G}_0 + G_2 G_1 G_0 + G_2 \bar{G}_1 \bar{G}_0$$

$$\underline{B_2} = G_2$$

$$\underline{B_1} = \bar{G}_2 G_1 + G_2 \bar{G}_1$$

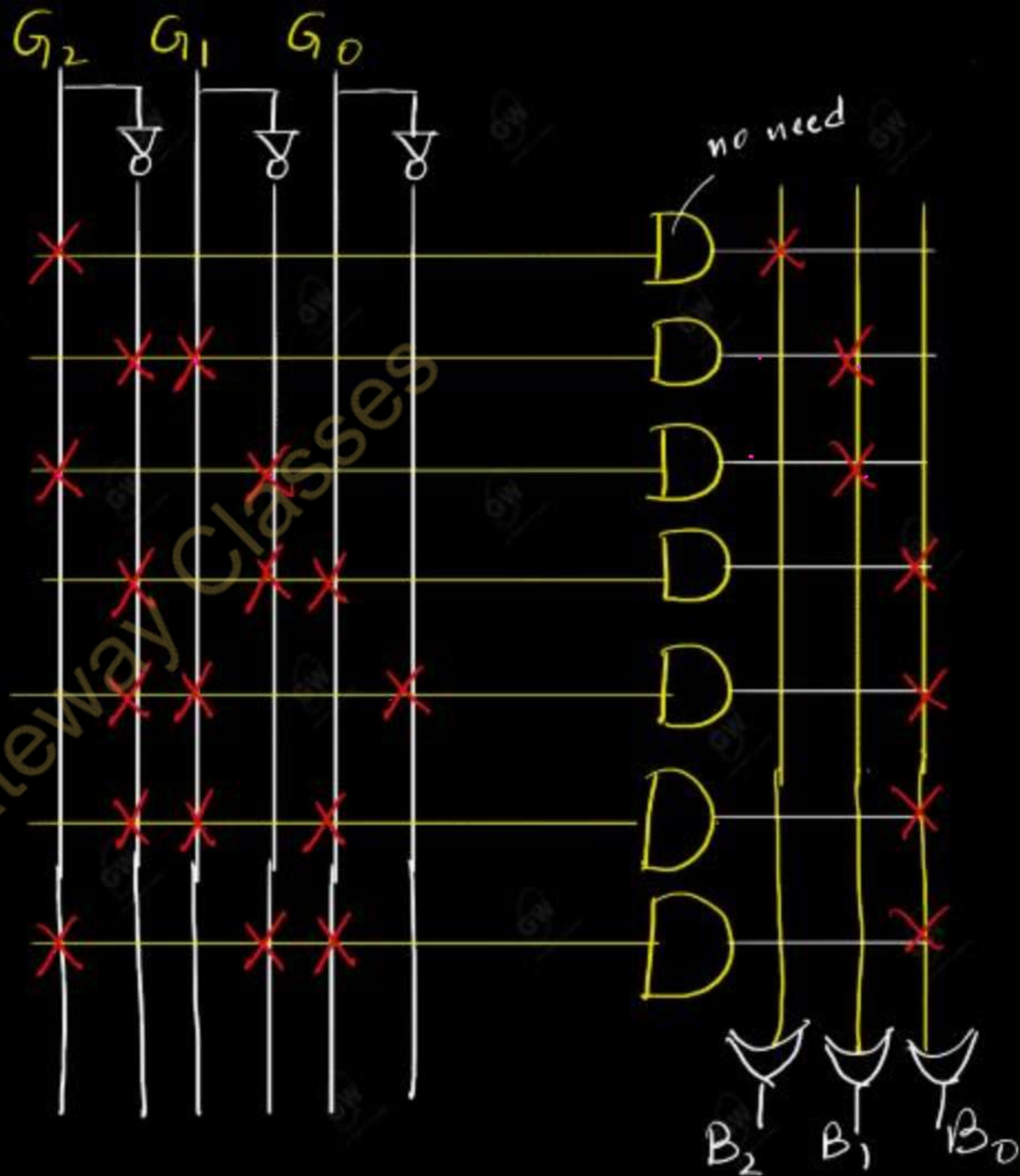
$$B_0 = \bar{G}_2 \bar{G}_1 G_0 + \bar{G}_2 G_1 \bar{G}_0 + \bar{G}_2 G_1 G_0 + G_2 \bar{G}_1 G_0$$

$$\text{AND Gates} = 7 - 1 = 6$$

(Total product terms)

$$\text{OR Gates} = 3$$

(Total Outputs)



Que - Implement BCD to Excess-3 using PLA

Concept

Excess 3 number = BCD number + (0011)

0-9

8 4 2 1

4 3

Gateway Classes

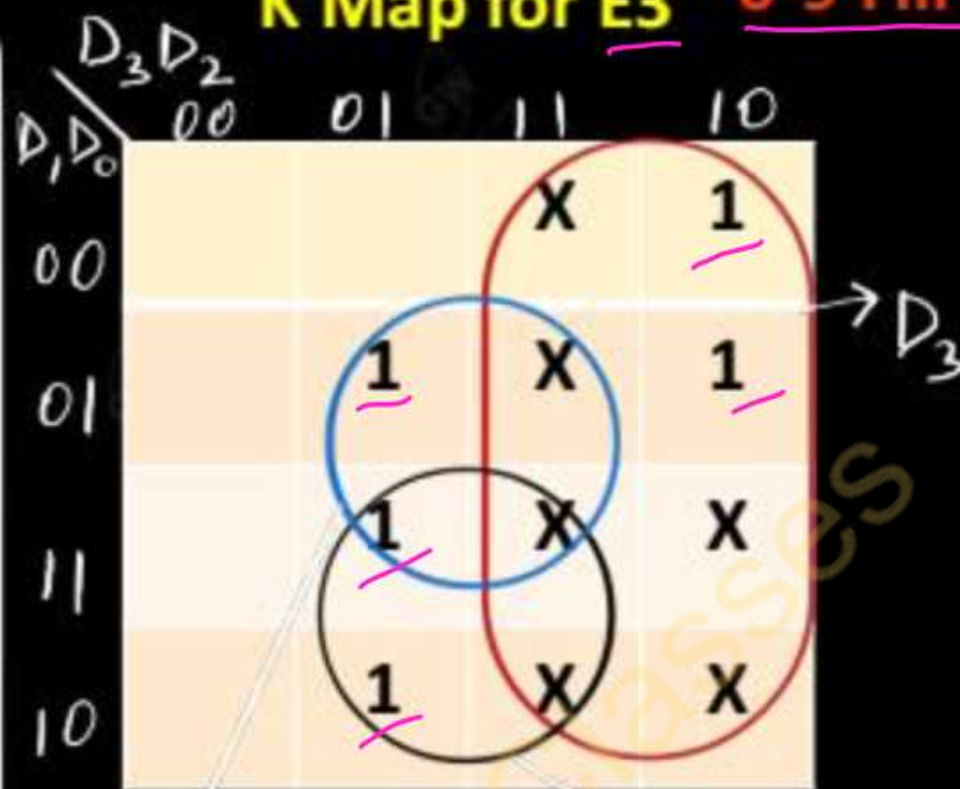
0-9

$$\begin{array}{r} 0000 \\ + 0011 \\ \hline 0011 \\ 0000 \\ + 0011 \\ \hline 0110 \end{array}$$

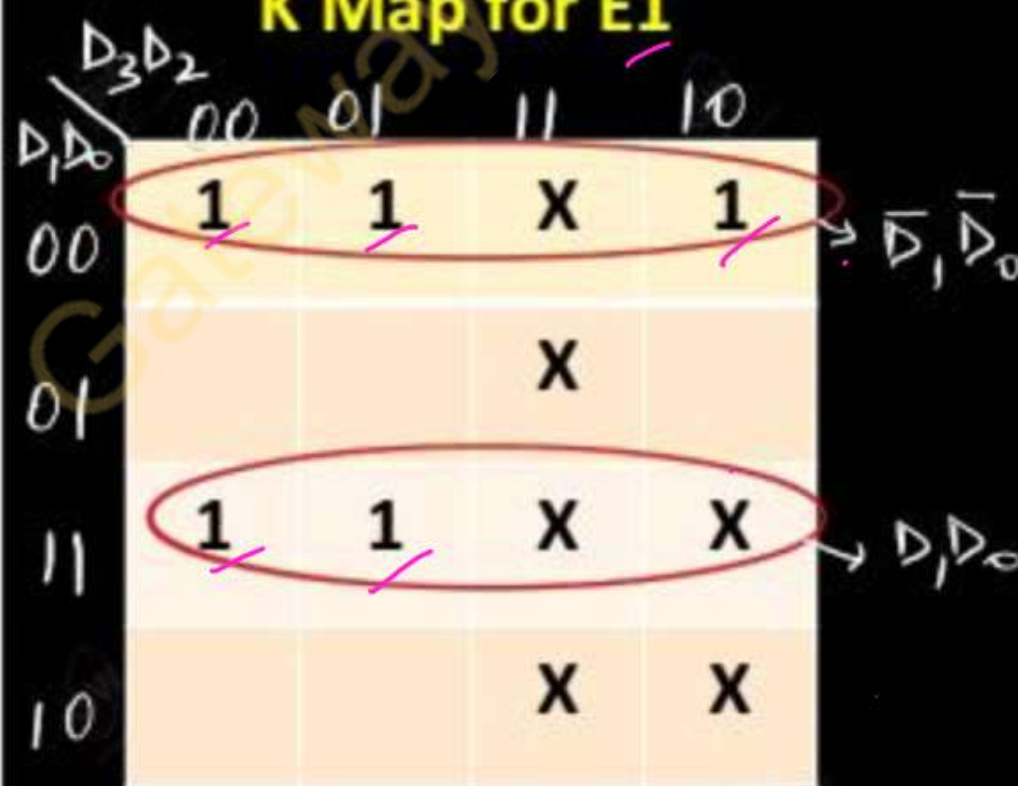
BCD INPUTS				EXCESS-3 OUTPUTS			
D3	D2	D1	D0	E3	E2	E1	E0
0	0	0	0	0	0	1	1
0	0	0	1	0	1	0	0
0	0	1	0	0	1	0	1
0	0	1	1	0	1	1	0
0	1	0	0	0	1	1	1
0	1	0	1	1	0	0	0
0	1	1	0	1	0	0	1
0	1	1	1	1	0	1	0
1	0	0	0	1	0	1	1
1	0	0	1	1	1	0	0

K Map for E3

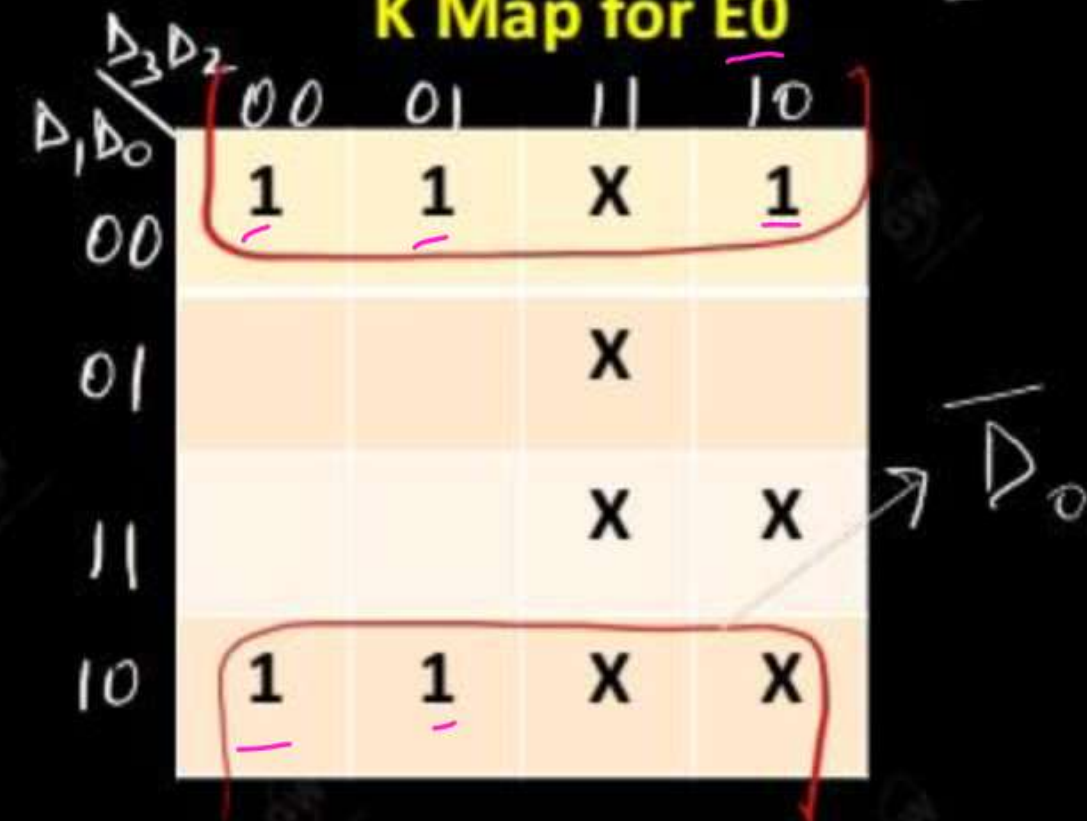
0-9 Fill 1, 10-15 Fill X



K Map for E1



K Map for E0



$$E_3 = D_3 + D_2 D_0 + D_1 D_2$$

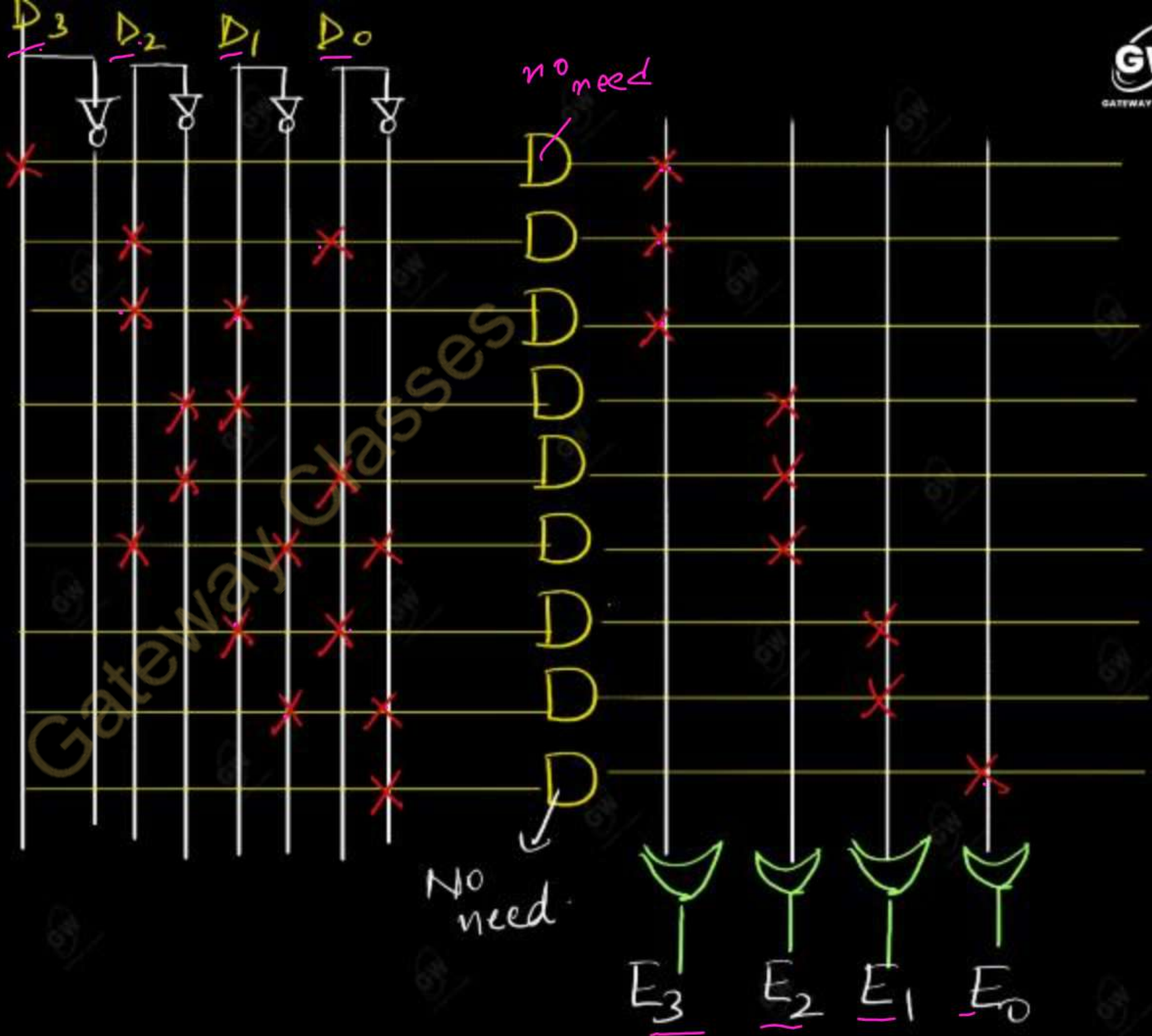
$$E_2 = \bar{D}_2 D_1 + \bar{D}_2 D_0 + D_2 \bar{D}_1 \bar{D}_0$$

$$E_1 = D_1 D_0 + \bar{D}_1 \bar{D}_0$$

$$E_0 = \bar{D}_0$$

AND Gates = $9 - 2 = 7$
(Total product terms)

OR Gates = 4
(Total Outputs)





Gateway Classes

Empowering Learners, Transforming Futures



Thank You



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